PROVISIONAL PATENT APPLICATION

Work-Compensated Fiber Communications System and Methods

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The disclosure concerns the complete communications system, not a separate software-verification invention. It describes a finite apparatus and proposed physical interaction constructions for material realization, fiber implementation and calibrated operation.

Prophetic-examples statement: The embodiments and examples in this disclosure are prospective constructions; no example is represented as an experiment performed or a result achieved.

Embodiment boundary. The actively guided work-compensated apparatus and O-BINARY-1 keep their own initial boundaries, 674 clock factors, resource accounts, decoder and error statement. CT1 and RCM-1 use their separately specified 65^3 inventory, contact law and phase instrument. A shared exchange-record interpretation does not identify their controls, combine their positive results, transfer an error budget between them or make the diagnostic a payload channel. RCM-1 uses source/calibration metadata for an explicitly nonpayload comparison; it does not relax the source-blind decoding boundary of E1. Each embodiment retains its separately stated operating domain. (See E1, E17 and E18.1–E18.3.)

CROSS-REFERENCE TO RELATED APPLICATIONS

This application belongs to a coordinated series concerning controlled circulation organizations, information systems, completion-scheduled operation and paired-endpoint accounting. The present subject is a source-modulated, work-compensated guided communications system with a fixed receiver and a separately maintained acquisition record. It does not depend on receiver-keyed instantaneous signaling, direct-adjacency transport, counterfactual concealment, or a source-independent replacement signal. The earlier archive-verification software is a supporting acquisition subsystem, not the subject substituted for this system. No priority claim or incorporation of an unpublished companion's essential disclosure is made here.

FIELD

Fiber communications; actively guided information transfer; relational angular encoding; distributed work-compensated exchange; source and receiver preparation; finite reference-supported analysis; failure-inclusive acquisition; and complete resource, timing, retained-state and repeat-operation accounting.

BACKGROUND

1. An information link requires a controllable choice at a transmitting station, a physical connection, a receiving operation and a mapping from actual receiver occurrences to recovered user information. A record containing the transmitting station's requested settings is not by itself evidence that the physical channel conveyed those settings.
2. A guide whose sites have different local energy tariffs must account for the difference during each transfer. A terminal-only energy balance can conceal work absorbed or supplied elsewhere. Similarly, distinguishable retained work histories can destroy the intended joint interference even when a position-only model appears coherent.
3. The present construction uses a tree-shaped guide and local work pairs. Each completed directed crossing exchanges the signal state and two complementary scalar work words. The unique route from the root to a vertex fixes its retained work word, allowing subsequent crossings to recombine without deleting the work record.
4. Complete operation also requires source/reference preparation, controller and clock resources, receiver pointers, retained unsuccessful outcomes and a declared mode for subsequent use. Returning the original source does not reset measurement records, consumed work or heat stock.
5. The apparatus below is an actively controlled circulation-cell fiber. It is not a claim that an ordinary glass fiber implements the specified contacts. Its first operating embodiment is finite and uses separately qualified fresh complete units. A continuous reusable embodiment includes an additional reset or replenishment construction.
6. Further embodiments specify transport, confinement, source and record operations with their respective contact laws, controls and retained accounts. Attenuation is funded by launched optical energy posted to ordinary accounts, not by a consumable-glass debit.

SUMMARY

A fiber communications system (100; FIG. 1) comprises a source-side information selector (110), a prepared source and retained reference assembly (120), a source-local angular modulator (130), a loader (140), an actively scheduled tree guide (200), a catcher (210), a fixed angular receiver (400), a receiver-occurrence decoder (500), local and independent completion controls (600), and retained work and acquisition assemblies (700).

The selector maps one of two user symbols to opposite source angular commands after common readiness. The receiver basis, guide program, endpoint settings and acquisition schedule remain fixed. The receiver decodes its completed pointer occurrence as zero, one or erasure. Transmitted command records, prepared-case identifiers and expected answers are excluded from its payload inputs.

Each guide edge has two complementary scalar work pairs. A positive exchange contact transfers the complete triplet signal between adjoining sites while swapping the work words so that the local energy and total angular momentum close. The work configuration is retained. Source return, local phase comparison and complete acquisition are additional physical operations, not substitutes for the source-to-receiver payload channel.

The disclosure separates the inherited finite apparatus, the new fixed-receiver binary protocol and alternative proposed contact compilations. Resource and error statements apply only to the version whose operations and assumptions support them. Fabrication, calibration and physical interfaces are selected for the contact prescription and operating domain.

Additional prospective constructions below specify reciprocal reusable transport, a curvature-producing routing branch, full-domain local-support classification, a quantity-priced joint-export conveyor, a same-law boundary obstruction, supplied-stock launch, scalar-fuel conversion, finite relative-phase generation and collection/record operations. Each construction retains its stated operating conditions and limits. They are alternatives or explicitly proved compositions, not interchangeable upgrades to O-BINARY-1. Reset and replenishment arrangements may be selected per deployment.

Exchange-record representation and memory-test disclosure. The governing description is a completed source–receiver exchange including guide, clocks, references and retained apparatus. A permitted reduction may forget only distinctions that no admitted future operation or requested record can reveal. E17 specifies the representation and its CT1 reductions. E18 specifies a separate same-bank repeated-contact discriminator with full-domain four-pointer scoring, independent error certificates, fixed bins, ten control rows and complete custody. Its persistent prediction is exactly 1/32; the accounted renewal alternative is more than 1/240 away. These embodiments retain their own controls and accounts separately from O-BINARY-1. (See E17 and E18.4–E18.9.)

BRIEF DESCRIPTION OF THE DRAWINGS

- FIG. 1 — Complete source-to-output communications system and separate audit channel.
- FIG. 2 — Tree-shaped fiber, backbone, transverse leaves and owned work supports.
- FIG. 3 — Work-compensated edge exchange and full-state complement.
- FIG. 4 — Source-local binary modulation with fixed receiver settings and retained reference.
- FIG. 5 — Receiver-only decoding boundary and excluded source-setting metadata.
- FIG. 6 — Finite attempted-symbol operation, timeout and unconditional retained record.
- FIG. 7 — Complete positive resources, retained outputs and fresh-unit repetition.
- FIG. 8 — Exact rate matching, guarded contact construction and material realization boundary.
- FIG. 9 — Layered paper support and prospective qualification gates.
- FIG. 10 — Alternative spectral compilation of the quadratic angular operation.

DEFINITIONS

1. **Fiber** — the elongated, physically connected, actively controlled circulation-cell guide defined here, including its backbone, transverse contacts, supports and work organizations. The term does not imply a glass core, optical carrier wavelength or passive refractive-index profile.
2. **Circulation dimer** — a complete nine-state local organization represented by two spin-one angular spaces, with singlet, triplet and quintet sectors of dimensions one, three and five. The imported representation is a mathematical tool, not a measured material identification.
3. **Tariff** — the same-account energy coefficient multiplying a specified local organization. The reference positive coefficient is κ; its SI calibration is separate.
4. **Scalar work pair** — two complete dimers with scalar words b=ss and a=(tt) of total angular momentum zero. A word transfer is retained physically, not removed from the account after a favorable result.
5. **Scalar bit** — four complete dimers with equal-tariff, zero-total-angular-momentum code words u=(a,b) and v=(b,a). All noncode complements remain in the apparatus.
6. **Common-coordinate matching** — an ownership relation in which the signal tariff at a support is κ/r and each owned work-dimer tariff is $\kappa/(2r)$, with the same stationary coordinate r. It is an exact type relation, not a stiffness approximation.
7. **Retained reference** — the named finite angular banks, partner/copy factors and scalar factors retained with the same original source; arbitrary reference correlations are not replaced after preparation or transfer.
8. **Completed occurrence** — a physical pointer or event state written by its specified completed operation, rather than a requested command or a software expectation.
9. **Payload decoder** — a fixed mapping from receiver occurrences to a user symbol or erasure, without access to the source command, prepared-case label or expected category.
10. **Erasure** — an attempted output not assigned a reliable binary value, including absent, incomplete, noncode, rejected or otherwise inadmissible receiver occurrences. It remains in the attempted-symbol denominator.
11. **Fresh complete unit** — one fully supplied source, guide, work, controller, clock, acquisition and retained-output organization qualified at the single initial boundary. A fresh identifier does not establish fresh physical stock or independence.
12. **Finite apparatus** — the construction configured with the declared material, initial-state, clock, control, thermal, readout and error requirements.
13. **Constitutive extension** — an explicitly proposed branch-level interaction or selection rule with its domain, complete action, accounts and independent failure conditions. It is not retroactively described as a deduction from existing premises.
14. **Record total variation** — the maximum difference in probability assigned to an event by two complete classical-record distributions. It is distinct from the error of an individual bit or the half-trace distance of retained physical states.

15. **Public comparison coordinates** — the independently specified geometric coordinates used to enumerate the apparatus. They are not meters until independently calibrated.
16. **Nonpayload audit** — the separate retained record of settings, completion, work, source return and failures. Its access to source settings does not authorize those settings as decoder inputs.

Quoted conventional labels identify mathematical representations or measurable phenomena and do not, by their use, establish a material ontology or a measured implementation.

Additional-construction notation. In E7–E16, Δ denotes a positive declared local tariff, τ a stipulated update or service duration, a a locally specified spacing, and $v_*=a/\tau$ only where that branch defines it. A conserved tariff is not automatically the temporal-translation generator; quasiphase is not an optical energy measurement. No $E=h\nu$ identification follows. CCK finite complex-state/unitary/Born calculus is the specified constitutive framework; spin representation, tensor, channel and spectral mathematics are imported tools. Each branch adds its printed contact, preparation, ordering and timing selections. Each embodiment includes accounts for intrinsic timing, joint action/tariff realization, geometry/support/recoil, physical preparation and readout. Indices and stock symbols are local to each construction and do not overwrite the inherited apparatus counts.

MATHEMATICAL AND CONTROL FRAMEWORK

Governing exchange-record description (E17). All descriptions below retain source, receiver, references, phase-bearing clocks and apparatus recipients inside the declared joint account. If POAMS is correct, retained glass/accumulator organization, propagating light and information share one root: the carry and the carried are the same, not separate cargo traversing an independent void, but optically projected relative turnings. For unit winding completed action is $\mathbf{h=2\pi\hbar}$, not energy or duration; unresolved relative amplitudes are not fractional completed objects. The question is which controlled source change produces a distinguishable receiver record relative to its clocks and references. Equality of current marginals alone does not license deleting a record. The general criterion is a representation discipline; E17's particular persistent-clock identity belongs to CT1 and is not imposed as a replacement law on E1–E16. (See E17.)

Full local state. Each dimer has $h=P_t+3Pq=2I+L_1\cdot L_2$, with energies 0, 1 and 3 before multiplication by its positive tariff. The signal includes all three triplet components. Singlets, quintets, wrong-work words, extra loading and noncode controls are not projected away.

Local transfer (FIG. 3). For a root-oriented edge $u\to v$, define

$$\mathbf{L_m=|t_m,s;\ bA,aB\rangle,\ R_m=|s,t_m;\ aA,bB\rangle,\ T=\Sigma_m|R_m\rangle\langle L_m|,\ K=(PL+PR-T-T\dagger)/2.}$$

K is a rank-three positive projector on the full 9^6-state six-dimer space. It is zero on the orthogonal complement. Pair A is owned at tariff $k_u/2$ per dimer and pair B at $k_v/2$; both complete pairs are mounted on the arrival side. The completed crossing has changes $\mathbf{(-k_u,+k_v,+k_u,-k_v)}$ in source, terminal, pair A

and pair B energy. It preserves all three total-angular-momentum components in the qualified matching domain.

Guarded constituent construction (FIG. 8). Order the complete dimers as (S_u,S_v,A_0,A_1,B_0,B_1). Define the full involution **F=SWAP(S_u,S_v) SWAP(A_0,B_0) SWAP(A_1,B_1)**. Let G_L=$P_t(S_u)P_s(S_v)$Pb(A)Pa(B), G_R=$P_s(S_u)P_t(S_v)$Pa(A)Pb(B) and G=G_L+G_R. The fixed predicates are P_t=h(3I−h)/2, P_s=(h−I)(h−3I)/3, Pb=$P_s\otimes P_s$ and **Pa=($P_t\otimes P_t$)(C−2I)(C−6I)/12**, where C=$(J_1+J_2)^2$ has triplet-pair eigenvalues 0,2,6. Pa selects the scalar two-triplet state; an energy-only predicate is insufficient. Since F G_L F=G_R, [F,G]=0 and **K=G(I−F)/2** reproduces the inherited contact on the complete space.

This is a prescription for one coherent guarded junction, not three sequential uncontrolled swaps through unpriced intermediate states. The proposed formed-junction law PCE-1 couples only the named guard-approved closure arrangements with the specified positive K, while retaining activation, reaction, outlet and controls. Its physical occurrence is an explicit type premise. Spin-component dependence, noncode leakage, missing work exchange, external-axis dependence, incoherent selection or failure of inverse crossing to restore work falsifies that premise. A stochastic swap with the same energy tally is a countermodel, not a realization of coherent K.

If independent work-pair rates produce **Δ=k_u+gB−k_v−gA**, then **‖[K,H_actual]‖=|Δ|/2**. Nonzero Δ lies outside the exact conserving domain. The rooted work configuration also has position-dependent shift **V(x)=−Σ(root→x)Δe**, producing OFF-dwell phase differences. A future tolerance or drive-compensation branch must close those accounts rather than treating a small rate mismatch as a free calibration allowance.

Phase-flat guide gate. The implemented gate is **exp(+iθP_t(u)) exp(+iθP_t(v)) exp(−2iθK)**. Its two endpoint tags are necessary. For a 32-bit base word w, the compiled positive-angle words are 2w,−w,−w modulo 2^{32}. Disabled and mixed-enable sectors retain their actual phases; they are not silently replaced with an intended code-space gate.

Tree work invariant. Let w(x) be spent on the unique root-to-x path and ready elsewhere. The map **J|x,m>=|triplet m at x; singlets elsewhere>⊗|w(x)>** intertwines each physical edge gate with a position exchange and the identity on the angular triplet. Reverse crossings restore the traversed work word. This proves joint signal/work coherence for the tree; tracing out work need not preserve bare position coherence. A looped guide is not covered without a separate path-consistency proof.

Source-local operations. With source s, frame f, P=$P_t(s)P_t(f)$ and A=$J_s\cdot$Jf, the positive angular contacts are **L=P(A+2I)** and **Q_2=P(2A^2+A−I)**. Their triplet-pair eigenvalues are respectively (0,1,3) and (5,0,2), and both vanish outside that sector. Finite positive symmetric sweeps over the retained frames implement the chosen angular command. The quadratic contact is supplied independently in the baseline. The alternative compilation in FIG. 10 does not inherit baseline resource or error totals automatically.

Binary version O-BINARY-1. The input alphabet sets the existing physical source command (source_setting0, source_setting1, source_setting2)=(0,0,b), selecting +Z for b=0 and −Z for b=1. +Z uses zero meridian Q_2 angle; −Z uses angle π. Both use zero subsequent axial L time. Receiver analysis is fixed Z with READ enabled, guide PASS, local endpoint blank off, and quiet FRINGE with n=q=reverse=0. The ideal receiver pointer is plus for zero and minus for one. None of the existing 821 registered cases pairs these two source choices under this identical downstream word. This is a new bounded control pairing, not the old campaign relabeled as a communications service.

In the reference-description triplet order (t+,t0,t−), Jx^2 has rows (1/2,0,1/2), (0,1,0), (1/2,0,1/2), and $(Jx^2)^2=Jx^2$. The meridian generator is Jx^2+I, so $\mathbf{\exp[-i\pi(Jx^2+I)]=2Jx^2-I}$, which interchanges t+ and t−. The finite-frame corrections and retained-reference mapping remain in the error account. A local trace-preserving source command leaves the reduced guide/receiver state unchanged before the data contact; that algebra does not excuse a command-dependent timing side channel.

Version-specific conditional bound. Both source commands are already admitted by the uniform angular-operation bound; the maximum meridian angle remains π. The same paid symmetric-sweep skeleton, full-triplet guide, scalar endpoint pullback, quiet endpoint padding, acquisition and initial joint boundary apply. No new physical classifier or exact input is inserted. The whole-operation telescope therefore retains coherent allowance $\varepsilon\le 0.096179707815\ldots$, together with $d_{in}=10^{-5}$, $q_{hot}\le 1.3490479445\ldots\times 10^{-7}$, $q_{halt}=10^{-4}$ and complete digitization allowance $d_{dec}=10^{-5}$. These assumptions cover the source-controlled operation, not an arbitrary external transducer.

For either intended b, every ideal complete record gives the source-blind decoder D=b. The event $\mathbf{F_b=\{D=1-b \text{ or } D=erasure\}}$ is therefore ideal-zero. Its coherent contribution is at most ε^2; the joint-input, nominal hot/halt and digitization perturbations enter linearly. Consequently, for this unchanged-device binary pairing,

$$\mathbf{P(\text{wrong bit})+P(\text{erasure}) \le \varepsilon^2+d_{in}+q_{hot}+q_{halt}+d_{dec} \le \delta=0.009370671101.}$$

The correct-output probability is at least 0.990629328899, conditionally. No separate smaller bound for wrong bits versus erasures is claimed. The full-record total-variation bound remains 0.096299842720; the predecoder same-factor density half-trace bound remains approximately 0.116203193748. These quantities are not interchangeable. The old campaign's compatibility threshold 0.019370671101 is not a measured or guaranteed communications error rate. Alternative contacts or a new interface require new accounts before this envelope transfers.

Finite resources. One inherited fresh attempt has 31,007,572,988,523 physical dimers and at most 25,577,373,557,351,935,515 native posts. The reference work and length units remain κ and a_0. Neither a bit rate in seconds nor a length in kilometers follows from those counts. Any newly introduced encoder, decoder, clock interface, repetition code or contact compilation has a separate overhead and error account.

SYSTEM ARCHITECTURE (FIG. 1)

Source and input (110–140). The prepared reference uses axial and meridian banks of 65,536 dimers each per bank, with primary census N=262,145, primary-plus-copy census 524,290 and 80 scalar M-register constituents. The complete source state is an invariant retained purification; its source marginal alone is $I_3/3$. Consequently, the declared relational receiver reference is essential, not an unpriced absolute axis. Source commands occur only after common readiness. Both binary choices share a padded horizon and launch address; no command-dependent early completion message is permitted.

The baseline factory uses its declared G word, 33 B/B† words, sixteen prepared-bank phases and a fixed amplitude-shaping sequence. Its finite mixed-radix clock and 6,449,012,345-bit two-positive-tape program are specified in the retained engineering annex. The tape is specified by that finite program. No second ideal source is appended after factory work.

Retained inventory	**Location and timing**
Axial/meridian reference banks sa/sm	Sender-local, installed before payload selection
Axial/meridian reference banks ra/rm	Receiver-local, distributed through the paid assembly before payload selection
Copy factors and original scalar M register	Retained source-side under the actual factory factor map
Guide work, receiver pointers and purifiers	Retained at their named installed/custody stations; not discarded or reset

The common seed is independent of the later bit. Predistributed reference does not itself transmit the bit: for a source-local U_b, **Tr_sender[(U_b⊗I)ρ(U_b†⊗I)]=ρ_rest** before the data contact.

Guide (200–210; FIG. 2). There are 6,144 backbone sites (0,0,z), with four transverse leaves (±1,0,z),(0, ±1,z) at each station and no leaf-to-leaf axial edges. This gives 30,720 vertices and 30,719 contacts. The source station is z=2,560 and the catcher station z=3,584, a separation of 1,024 edges. Each support owns the signal and two work dimers for each incident edge. Exact common-coordinate matching makes its conserved load $\mathbf{N_V=h(signal)+(1/2)\Sigma h(owned\ work)}$; the good packet has $N_V=1$ while failed sectors keep their actual loads.

The loader prepares 512 cells with $\mathbf{f_j=\sin^3(\pi j/513)/\sqrt{(5\cdot 513/16)}}$ and positional spinor $(\sqrt{0.7},i\sqrt{0.3})$. It uses 1,023 adjacent positive rotations. The guide performs 1,280 updates, each comprising backbone even and odd matchings at asin(2/5), a leaf phase $\pi/2$ and four nonzero transverse matchings of angle 10^{-6}. A translated inverse loader catches the packet. All 2,049 32-bit guide words are fixed independently of the payload. The two positional components are not renamed polarizations.

Receiver (400; FIG. 4). The fixed Z receiver uses the finite axial bank and a scalar phase instrument. Its two-bit phase circuit, classification and actual inverse retain the angular source and workspace. The completed pointer pairs are W=v with $R=r_0=(a,b,b)$ for plus, $R=r_1=(b,a,b)$ for minus and R=rr=(b,b,a) for

reject. W=u with r_0 denotes omitted READ and is not a decoded zero. All noncode and third-component outcomes retain their defined classification.

Endpoint and source return. A full equal-tariff 9×9 SWAP transfers the actual receiver source, including its retained correlations, to the phase endpoint. It does not insert a newly prepared replacement. The full phase protocol is first mixer, acknowledged tags, optional quadrature and recombiner. Its right-output probability is:

$$\mathbf{pR = [1 - \cos((-1)^{reverse}\, n\pi/8 + q\pi/2)] / 2.}$$

This is local endpoint modulation after arrival and is not the remote payload channel. At the witness kB=7κ/8, terminal energy plus net work equals 7κ/8+κ/8=κ. The original source returns under priced custody operations; return does not reset the complete apparatus.

Decoder (500; FIG. 5). The concrete binary version waits for the inherited final seal and reads 71 actual observer scalar occurrences derived from receiver/clock outputs: three W/R output bits at observer dimers 668–679; W/R-present at 680–683; final seal at 684–687; a 65-bit accepted OFFER latch at 3042–3301; and latch-present at 3302–3305. The whole acquisition remains intact. Exact physical kit/slot identity binds the occurrences but never chooses a bit.

The reducer requires valid scalar words, W/R-present=1, seal=1, latch-present=1 and an actual LSB-first OFFER count strictly below $\mathbf{16P^{*}+4=24,072,822,171,625,280,596}$. Here $\mathbf{P^{*}=1,504,551,385,726,580,037}$. Printed W/R bits 001 decode to zero and 010 to one. Every other three-bit word, missing/partial/noncode occurrence, wrong identity, absent capture, unsealed output or late/tied OFFER yields erasure. In particular, a valid 010 received after intended zero is a wrong bit, not an erasure detectable by consulting transmitter truth.

This embodiment provides source-blind **post-acquisition output** at the specified acquisition boundary. Its inputs exclude source settings, prepared-case IDs, endpoint labels, expected answers and the original archive classifier's forbidden flag. A separate complete-record audit may inspect those fields, but it cannot select, repair or supply the payload bit. The semantic reducer specifies the digitized-word map. A physical occurrence-to-digital transducer is selected to satisfy the one complete d_dec allowance. An earlier local-pointer interface uses its own timing and qualification account.

Completion and records (600–700). The inherited apparatus has one initial joint nonclock/reference half-trace bound of 10^{-5} and exactly 674 independent pure clock/control factors. All source work, new supports, heat, control, record and purifier factors belong to that initial boundary. There is no new perfect inlet between stages. Native operations include full positive switching, acknowledgment, fail sectors and bounded progress. UNA-65 supplies one atomic 65-bit capture on 207 named scalar operands. Sequential bit sampling is not an equivalent operation. The clock seals failures as well as successes. The protocol reserves 1,811 named scalar output occurrences on 7,244 dimers; absent, malformed or uncompleted observations remain unavailable failures.

Payload-independent observer metadata. O-BINARY-1 fixes the legacy 21-bit prepared-case word at observer dimers 520–603 and the five-bit source-stratum word at 604–623 to valid constant zero before the source bit is selected. Both remain inert legacy metadata and are never rewritten to reflect b. They do not feed native receiver controls, readiness, OFFER or the decoder; the observer's outgoing launch remains source-independent. Thus the exact-C initial organization is common to both payload values, not an early bit-dependent channel. The old archive software's source-stratum-versus-case comparison is not a native guard and is not an O-BINARY validity rule. A new O-specific audit must not invoke that obsolete semantic comparator or rewrite payload outputs from transmitter truth.

OPERATING METHOD

1. Allocate one complete fresh unit and its fixed receiver, guide, endpoint and acquisition programs. Declare the initial joint-state, material, thermal, clock and decoder domain before any source preparation.
2. Perform the complete source/reference factory and verify readiness through the declared causal controls without success-selecting away attempted failures.
3. Apply the selected user symbol only at the source-local angular modulator after common readiness. Keep receiver settings and the full launch horizon fixed.
4. Load, propagate and catch the same signal using the fixed guide program. Retain the work configuration and all failed or incomplete sectors.
5. Operate the fixed receiver and retain its completed physical pointer occurrence. Do not infer an early electronic output before an actual transducer and its timing have been supplied.
6. Complete the specified source transfer, quiet endpoint operation and source return. Retain the work, thermal, control and measurement histories rather than treating a returned source as a reset.
7. Perform the bounded acquisition and unconditional seal. At that boundary, apply the source-blind 71-occurrence reducer to output zero, one or erasure. Store missing, late, duplicated, noncode and adverse outcomes in the attempted-symbol record.
8. Keep payload output and subsequent source-informed audit distinct. Where the record is acquired only after later collection, label that audited output accordingly rather than claim an earlier receiver delivery time.
9. For another attempt, use a separately supplied and qualified complete unit. Introduce reuse only under a separately constructed replenishment/reset law.

The source-to-pointer operation, native-to-digital conversion and final audited record can have different availability times. A whole-program post count is not automatically a receiver latency. The final observer cursor is **C=24,072,822,171,625,355,478**, distinct from the sum of all work posts. With independently qualified counter-advance time bounds τC^- and τC^+, elapsed origin-to-seal time lies between $C\tau C^-$ and $C\tau C^+$. The qualification must cover the whole finite horizon; an average oscillator rate alone is insufficient.

For N information bits and odd repetition r, allocate Nr complete physical units. The fixed sequence of unit addresses is independent of the data. If separately qualified setup and auxiliary interfaces have bounds t_setup+ and t_aux+, a conservative sequential frame duration is:

T_frame+ = t_setup+ + Nr(CτC+ + t_aux+).

This is a conditional parameter relation, not a numerical bandwidth claim. Guide station separation is **$1{,}024a_0$**, not an asserted overall package size or a generic constant-performance distance law.

Fresh-kit lifecycle embodiment. A corrected-kit energy configuration uses K=35,454,990,850,307 dimers per attempt in its own scope. It is not a replacement for the version-specific inherited count above. De novo SEED manufacture includes its own resource account, and source return is distinct from renewal. This embodiment supplies fresh whole kits. The additional reciprocal/export constructions below use separate contact laws and do not change this apparatus's reset, cadence or O-BINARY-1 resource counts.

LAYERED SCOPE

- **O0 — inherited finite apparatus:** the exact source, guide, same-source endpoint, complete accounts and acquired-record construction under its supplied types.
- **O1 — binary system protocol:** source-local +Z/−Z modulation, fixed Z receiver and source-blind output mapping; a new protocol derived from the source operations, not the 821-case campaign relabeled.
- **O2 — proposed contact construction:** guarded local permutation and alternative angular-contact compilation, with separately stated constituent interactions, full complements, costs and material implementation requirements; additionally E7–E16 disclose separately versioned prospective operations and scoped obstructions, without combining different laws' results or transferring the O1 bound.
- **O3 — physical input/output and calibrated fiber embodiment:** actual supplied types, source selector, receiver transducer and independent work/length/time calibrations.
- **O4 — repeatable engineered product:** selected and qualified reset/replenishment, tolerances, operating-rate limits and finite-resource improvements. This layer is not inferred from O0–O2.

Seed-factory resource-allocation embodiment. At the fixed tolerance, a de novo seed-factory allocation comprises 2.989759102×10^{53} dimers, 3.502193518×10^{52} posts and 6.601479256×10^{53} service weight. The first two allocations are 8351.66×/8351.98× below the respective first-compiler allocations. The compulsory analog row-carry allocation exceeds 3.29282×10^{31} service norm and must be accounted separately from an additive certificate. The upper-allocation-to-corrected-kit ratio is **$8.432547943498523\times10^{39}$**. These quantities specify upper allocations, not minima or optima. Distributed

returning/interleaved and state-specific configurations may be selected per deployment with their own complete service accounts. The supplied-primitive G/B/D and corrected-kit/transport configurations retain their separate counts, with no double counting of transport improvements.

Reset and replenishment selections. A deployment may select finite matched isolated ready-stock inlet/outlet exchange (FLRH-1), nonwrapping work ladders and a separately specified cadence law. The inlet/outlet arrangement retains old-state recipients so that reset preserves injectivity and records. For a cyclic work ladder, $w=0\to M$ raises the work account by $M\delta$ instead of lowering it by δ; such a selection requires a separate $(M+1)\delta$ account. A selected clock is configured to satisfy the required commutator relation. At a handoff tolerance $\delta=10^{-5}$, an imperfect re-entry contributes $\delta+(1-\delta)\sin^2(\varepsilon/2)>\delta$; protected support, stronger supplied preparation, recovery or exact-native handling is selected to satisfy the next-input bound. A halted-ON branch includes priced isolation, rescue and re-enable operations. An 84-row custody cover totaling K may be partitioned into invariant, restorable and archival stock for the selected deployment; the corresponding restoration inventory R, comparison $R<K$ or $R\geq K$, I_R and breakeven are determined for that partition. Capacity upper bounds remain distinct from consumption.

Additional guide selections. A vector-model embodiment may use Maxwell-equivalent transport with separately specified finite spatial realization and constituent, specimen, support and outlet construction. For the fixed K/P_t guide after export, spent-work second injection lies in the edge kernel even at equal tariffs; P_t tags add phase only and the tree provides no bypass. First active delivery retains its stated action. Repeated injection therefore uses restored work stock or a separately specified guide law.

QUALIFICATION GATES

- **O1 payload separation:** a withheld source symbol affects the actual receiver pointer while receiver commands, launch schedule and decoder input schema remain unchanged;
- **O2 source/reference custody:** retain the original source, actual reference factors, work and recoil through preparation, guiding, analysis and return;
- **O3 local operation:** establish the complete contact action, including singlet, quintet, wrong-work, noncode and multi-occupation complements;
- **O4 matching:** establish the actual common-coordinate ownership relation; independent mismatch is not admitted as a harmless calibration constant;
- **O5 no leakage:** exclude case identifiers, source-setting fields, early command-dependent traffic and expected-answer tables from the payload decoder;
- **O6 readout:** derive and qualify a physical occurrence-to-output transducer, with no synthesis of observations from requested settings;
- **O7 attempted outcomes:** retain timeout, blank, rejected and malformed outcomes in the denominator, and separate bit error from erasure;
- **O8 complete resources:** retain positive inlet/capacity, support, activation, heat, record and return accounts, including halted sectors;

- **O9 versioned bounds:** recompute a bound for changed contacts, schedules or data fields rather than inherit an earlier certificate by naming similarity; and
- **O10 repetition:** supply independently qualified whole units or a justified per-history conditional record law before using concentration or independent-block coding claims.

The selected embodiment applies these qualification operations to its own complete state, controls and output records.

Additional operating conditions. O8 includes the selected seed-factory and analog row-carry resource accounts. O10 includes the selected renewal, re-entry, isolation, clock and work-ladder arrangements. Each embodiment has its own performance, capacity, bit-rate and efficiency account. All failed operating outcomes remain in the original attempted-slot or block denominator; neither heralded success nor a reduced marginal substitutes for the complete output/reference account.

Transport embodiment selections. A transport configuration specifies: (1) useful channel and complete traffic/reference quantifiers; (2) every carrier/support/router/delay/output sector; (3) one actual full-domain contact/succession and source/receiver circuit; (4) coherent routing topology, memories and outlets; (5) causal decoder, finite latency/cadence and exact drain or a prospectively changed error target; (6) fresh stock, opening inventory, source/return/price/custody counts; (7) same-law primary/complement mode census, primary plane versus complete transverse law, and curvature; (8) independent a/τ, band and base/refinement calibration; (9) full-load, cross-time-reference, retained-memory, lost-coherence, stock and uniform/zero-routing controls; (10) the complete operating conditions of its selected law. A hybrid uses an explicitly specified combined law; RAR45 curvature is not assigned to unchanged DEX4.

Confinement embodiment selections. A confinement configuration specifies: (1) changed structural class, target and full quantifiers; (2) full core/cladding/exterior/interface/support/reference inventory and preparation; (3) one complete law/order/control and scope of any new bulk transport operation; (4) physical contrast/geometry and derived interface/reaction map; (5) full spectral/localization certificate, exterior sectors, tails/leakage and non-guided outcomes; (6) same-graph zero contrast, widths, finite span, endface/drain/export and controlled macroscopic limits with bend/strain/disorder/radiation exclusions; (7) full-reference port/rail access, causal decoder and recipients; (8) opening stock, reset/drain/export and separate physical preparation/control/clock/action/tariff/work/heat accounts; (9) independently frozen contrast-removal, exterior, off-code, support, width, rail-fault and edge discriminators; (10) complete custody records, configuration readback and a predetermined acceptance or rejection rule. More disconnected lanes, axes, tariff labels, source codes or a chosen profile do not supply confinement.

Source embodiment selections. A source configuration specifies: (1) changed target and full source/circuit/fuel/blank/reference/controller/support inventory and initial correlations; (2) actual full-domain pump, fuel/blank renewal, relaxation if claimed, service/control ordering and failures; (3) independently bounded shared clock/action/energy specification, temporal generator versus tariff and switch/recoil reactions, preserving the old L0 exclusion unless a new launcher is frozen; (4) finite phase-bearing mechanism, across-slot/block coupling, degradation recipient and restoration/interblock-lock law;

(5) duty/capacity/stability/noise/horizon/transient/exhaustion domain; (6) every-attempt complete state and physical accounts including compulsory partner, purifiers, reflected/unguided/noncode outlets and final drain; (7) joint consequences from the same maps, not fitted desired outputs; (8) independent calibrations and no-pump, blank, unready, independent-stock null, correlated-stock witness, wrong-phase and hidden-reset controls; (9) exact full port/dependency, same-law guided loader if claimed and all rejected channels; (10) preparation records, fixed configuration and readback, separate pump/phase/composition gates and stopping disposition. A desired density, G, linewidth, debit or interface name is not a construction; source coherence alone does not supply record decorrelation.

Record embodiment selections. A record configuration specifies: (1) genuinely changed target/quantifiers, useful record, error and horizon; (2) achievable full joint traffic preparation including source, medium, controller and future-accessible references; (3) full scalar/triplet/quintet/pointer/bias/service/stock inventory; (4) one lawful full-domain contact/write/transport/storage/export update with occupied/nonfresh complements; (5) same-law arrival/collection channel and separate circuit/gain identification if claimed; (6) actual storage/isolation/readout organization, disturbance domain and accounts; (7) named correlation recipients and proof of causal inaccessibility when asserted; (8) finite horizon, capacity/exhaustion, all-attempt denominator and priced reset or explicit one-shot stock; (9) independent timing/action/tariff/work/recoil/heat/pricing boundary preserving the D0 exclusion; (10) predetermined null/correlated-factorized/loaded/isolation/reuse discriminators and all inherited conditions. Accessible future correlations remain retained unless the specified operation exports them beyond future causal access.

EMBODIMENTS

E1—finite binary fiber system. The source selector chooses +Z or −Z; the receiver performs fixed Z analysis with READ. PASS and the quiet endpoint word are fixed. Every attempted symbol consumes its full allocated unit. The native output is the receiver pointer; the concrete semantic digital output is its source-blind 71-occurrence reduction after final seal, not the globally informed archive classification. The embodiment includes a physical input/output transducer. This is the main communications embodiment.

E2—complete acquired-record system. E1 is accompanied by causal completion events, actual token and source diagnostics, atomic clock capture, unconditional seal and all-attempt reduction. A wrong-valid observation may remain diagnostically visible while also counted as forbidden by the audit. Acquisition-invalid records remain unavailable. The audit is not the payload channel.

E3—structured circulation-cell fiber. The guide comprises the specified comb of full dimers, correlated supports, local positive contacts and mounted work pairs. Member geometry, ownership, port identity, positive activation, return route and retained heat organizations are fixed. A cross-section drawn as a fiber jacket supplies none of these operations by itself.

E4—guarded exchange contact. The constituent construction in the mathematical framework gives F, G and K=G(I−F)/2 on the entire six-dimer space. The joint guarded contact must implement that coherent positive exchange, not sequential uncontrolled swaps. Its projector specifies the full action and complements; the formed-junction structure is configured to select that coherent interaction.

E5—alternative angular-contact compiler C (FIG. 10). On the complete 81-state dimer pair, $\mathbf{Q_2=2L^2-7L+5P}$, where $P=P_t(s)P_t(\text{frame})$. L has eigenvalues 0,1,3 inside the triplet pair and zero outside it. P is essential: the two zero-eigenvalue sectors must not receive the same Q_2 phase. Supply five scalar bits: predicate f, phase register y of two bits, and temporary flags r_s,r_f. Their nominal input is f=0, y=00, r_s=r_f=0, supplied within the one joint input/preparation bound; it is not a new perfect inlet. They comprise 20 bare dimers before factories, control, mounts and heat.

To evaluate P coherently, use the inherited presence interaction $\mathbf{\exp(-i\pi P_t\otimes D)}$, with $D=(|u\rangle-|v\rangle)(\langle u|-\langle v|)/2$, to write the two actual triplet predicates to r_s,r_f. Apply scalar Toffoli from those flags to f, then reverse the two writes. This gives f→f XOR P and restores the flags, at a cost of four presence gates and one Toffoli per P evaluation. It is not a software test supplied with a hidden sector label.

For an intended $\mathbf{\exp(-i\tau Q_2)}$, execute in order: (i) evaluate P into f; (ii) apply positive-sign F4 to y; (iii) apply the two controlled blocks forming $\exp(+2\pi iyL/4)$, namely $\exp(+i\pi L/2)$ and $\exp(+i\pi L)$; (iv) apply F4† so y holds the exact L eigenvalue; (v) apply $\exp(-5i\tau)$ only at (f,y)=(1,0) and $\exp(-2i\tau)$ only at (1,3); (vi) reverse the spectrum-resolution sequence; and (vii) uncompute P. All other phase addresses have unit phase. F4 has entries $\exp(2\pi ijk/4)/2$ for j,k=0…3. Nominal scratch returns exactly, and the data receive the stated Q_2 evolution including identity on their non-triplet complement. A source-controlled version enables only the addressed phases and runs the same padded spectroscopy/uncompute sequence for both symbols.

Each scalar primitive has its full code-projector extension; their finite product determines wrong-scratch and noncode behavior. Those failures are not a new perfect inlet. Positive inverses exist for the local integral L spectrum: $\exp(+i\pi L/2)=\exp(-3i\pi L/2)$ and $\exp(+i\pi L)=\exp(-i\pi L)$. This is not extended to the receiver's noninteger collective spectrum. Scalar phases and their positive switching supplies remain paid.

Each local Q_2 replacement uses four controlled-L blocks, four F4/F4† blocks, eight presence gates, two Toffolis and two addressed phase blocks. The inherited symmetric sweep has n(2K−1) local operations; with K=65,536 and n=512, the fixed padded source-plus-analyzer catalogue has **134,216,704 replacements**. Thus there are 536,866,816 controlled-L blocks and the same number of Fourier blocks, plus 1,073,733,632 presence blocks and 268,433,408 Toffolis. One stationed scratch bank can be reused only through the specified coherent uncompute and paid routing. Fresh scratch per replacement would require 2,684,334,080 bare dimers, not the complete apparatus bill.

With full-space per-block errors εL, εF, εpresence, εToffoli, ε0 and ε3, the additional telescope is bounded by $\mathbf{NQ(4\varepsilon L+4\varepsilon F+8\varepsilon presence+2\varepsilon Toffoli+\varepsilon 0+\varepsilon 3)}$ before the newly accounted preparation, routing, thermal, clock and acquisition contributions. The selected implementation integrates physical footprint,

block cadence, native service, new heat opportunities, positive capacity and final horizon into its account. C removes an independent Q_2 primitive under its remaining assumptions, but it is not the performance version E1 and does not inherit its δ or its total resources.

E6—fresh-unit block. A finite message allocates one complete unit per attempted symbol, or r=2t+1 complete units per repeated information bit. The block decoder outputs a value only when more than r/2 of the original r slots give that value; otherwise it outputs erasure. Erasures remain in the denominator. Noncorrect block output requires at least t+1 noncorrect physical symbols. Under independently qualified whole units, or the per-history bound $P(F_i|\text{complete prior history,current b})\le\delta$, its upper probability is **$f_r=\Sigma(k=t+1\ldots r)\ C(r,k)\delta^k(1-\delta)^{(r-k)}$**. For r=3,5,7 the respective bounds are approximately 2.61783×10^{-4}, 8.11312×10^{-6} and 2.63846×10^{-7}. For N information bits, any noncorrect bit has probability at most min(1,Nf_r) by a union bound. Without the whole-unit or per-history premise, shared faults can defeat these improvements. Physical grouping/decoder logic and external interfaces have additional costs and failure allowances. A block is not an indefinitely reusable stream.

E7—prospective reciprocal reusable transport law CRP2. This is a different supplied law from the spent-work K/P_t guide. In its base open 8×8×8 organization, each vertex has twelve full nine-state dimers: six directions d and two storage stages r=0,1. A retained scalar support G fixes adjacency, spacing and its invariant state. The complete reciprocal contact W exchanges all 81 basis pairs of any two dimers, independent of eligibility, occupation, purity or work words. With $h=P_t+3P_q$, the declared account is $H=\Delta\Sigma h+E_G I$. One stipulated update is **U=FBDR**, rightmost first, with $R=\exp[-i\tau\Delta\Sigma h/\hbar]$; D exchanges the two stages on site; B exchanges opposite-direction stage-zero seats across each internal axial bond; F exchanges the opposite stage-zero directions on site. Missing B bonds use full-state boundary SWAPs with external seats; outgoing old seats are retained/exported. Fixed order and duration are constitutive selections, not a controller-free material implementation or the deletion of paid controllers from E1.

On the frozen $p\in[0.01,0.10]$ independent-input interval, and on the full angular/reference and multiple-occupation domain of the exact permutation, the base law gives **probability-one complete angular/reference transfer with zero channel error after 16 constitutive update cycles**, uniformly for every finite traffic duration without exhausted eligibility/purity/stock. The known scalar gap phase is retained. A complete drain supplies 6,144 external scalar inputs and retains/exports 6,144 old factors rather than erasing them. This is a conditional reuse theorem, not a new communications-service performance claim. The same law has the optical constraints: **zero dispersion curvature, no law-derived transverse-only sector, $v_g=v_*/2$ and $n_g\ge2$** under its frozen $0<v_*\le c$ calibration, excluding the reported approximately 1.468 in that domain. A selected source code or fitted velocity does not change these constraints.

Failure conditions include any in-domain full-load/reference transfer error, consumed or altered unlisted support, failure to retain drain outputs, nonconserving full contact, or a physical support/recoil/timing/action account absent from the model. Optical curvature, transverse structure or

index outside the frozen predictions rejects the candidate as an identification, not observed fiber. Its reuse and optical properties apply together. No E=hν inference follows from its tariff or eigenphase.

E8—prospective curvature branch RAR45. Use six full nine-state dimers per vertex in the base open 8×8×8 organization, one for each signed axis, and the invariant scalar-support premise. On a complete pair let $\mathbf{C_\theta=\exp[i\theta(I-W)]=\exp(i\theta)[\cos\theta\, I - i \sin\theta\, W]}$, with full SWAP W, not a one-excitation-only rule. Apply this contact to each on-site opposite pair, then the same kind of internal/boundary full-state exchange B and on-site direction reversal F: **U=FBCR**, R=exp[−iτΔΣh/$\hbar$]. The base independent branch selections are θ=π/4 and Δτ/$\hbar$=π/8. The complete law conserves its declared tariff and all three angular components; all loaded, quintet and reference sectors remain. A stationary product-density witness and bounded inventory do not establish useful transfer.

The conditional spectrum has **nonzero normal/anomalous curvature**. At the frozen base, $\mathbf{|v_g|\le v_*/\sqrt{2}}$, and the shape band permits $\sqrt{2}\cdot c/v \le n_g \le \sqrt{3}\cdot c/v$. The reported **1.468 is admissible, not predicted**, only when independent calibration gives $\mathbf{v_*/c \ge 0.9633607373}$, within $v \le c$. *The √2 bound is base-only; refinement member r has* $|v_g| \le v\cos(\pi/4r)$. No target-fitting, finite-open optical packet result or material identification follows.

Finite-latency and drain limits. Exact single-output angular/reference delivery fails at every finite latency in the full-load witness and stated stationary bath; residual information survives every finite permitted blank drain; longitudinal and transverse components propagate identically. A stationary density, asymptotic single-excitation drain, or a favorable bulk derivative cannot replace exact finite-useful delivery/drain. Those are decisive failure conditions for this posed target, in addition to any omitted full-space reaction, changed support, lost reference or unsupported physical timing/energy realization. RAR45 does not inherit CRP2 reuse, and CRP2 does not inherit its curvature.

E9—prospective local-component classification N1c. For one full carrier $C=V_0\oplus V_1\oplus V_2$ and a spin-one support S in pure nematic preparation |1,0>, product with arbitrary carrier/reference/record inputs, the full 27-state decomposition is $\mathbf{V_0\oplus 3V_1\oplus 2V_2\oplus V_3}$. Angular conservation permits all **U(1)×U(3)×U(2)×U(1)** multiplicity blocks. The nondegenerate additive tariff $H_C=\Delta(P_1+3P_2)$, $H_S=\varepsilon I$ reduces them to **(a), diag(r,b,d), diag(c,e), (f)**, with unit-modulus phase entries. Here r labels carrier-spin-zero/total-spin-one; a,b,c the carrier-triplet total-spin-zero, one, two sectors; d,e,f the carrier-quintet total-spin-one, two, three sectors. All multiplicities and cross-sector coherences remain.

The fixed D_DIR triplet directional criterion holds **iff not(a=b=c)**. Exact support return, including decorrelation from every carrier/reference/record, holds **iff a=b=c and d=e=f**. Thus the **directional-and-exact-catalytic conjunction is EMPTY** in this frozen class. This is mutual exclusion, not a claim that every nondirectional member is catalytic. The returning family is exactly $\mathbf{(rP_0+bP_1+eP_2)\otimes I_S}$; it is direction-blind and reusable for every finite encounter sequence and jointly entangled input. A directional reduced channel has **Choi rank three**, so a pure-environment dilation retaining that fixed channel needs

a complementary output of dimension at least three. The full outgoing support, every q,q′ coherence and all references must be retained. The sufficient fresh-stock bookkeeping protocol supplies and exports one full support per encounter, with Kε in/Kε out; that tariff tally is not a preparation source or physical price.

Any extra sector, nonunitarity or nonzero total-angular/tariff commutator violates the class; impure, correlated or payload-dependent support preparation violates its input premise; unchanged support means or density alone do not pass exact product return. An omitted correlation recipient or unpriced stock replacement fails the export account. Enlarging a hypothesis is a new posing. No universal finite-reference/passive-media no-go, spatial law, spectrum or fresh-reference manufacture is established; CRP2/RAR45 are nonmembers.

E10—prospective priced-export conveyor DEX4. Exact local support catalysis is deliberately dropped. A complete packet is $\mathbf{B=C\otimes S_1\otimes S_2\otimes S_3\otimes S_4}$, with full nine-state C and four spin-one supports, dimension 729. Each support is prepared from an independently supplied pure nematic token by a full matched SWAP; the old raw-support state and all its correlations leave on the stock-token output. This transfers supplied purity/orientation, not its manufacture. Four whole-packet guide seats and a receiver service slot have fixed inventory. Four opening packets are supplied separately. Scalar control and timing records, old-state outputs and all terminal custody are explicit external streams.

On $C\otimes S_j$, supply $\mathbf{U_j=I-2\Pi_{carrier\ j=1,total\ J=0}}$, identity on the other sectors. In a Cartesian triplet basis the selected scalar is $(|x,x>+|y,y>+|z,z>)/\sqrt{3}$. Each tick applies the respective contact to each guide seat and advances whole packets by the fixed full-factor port permutation. Let $V_4=U_4U_3U_2U_1$; the explicitly paid receiver applies $\mathbf{D=V_4\dagger}$, using the complete four support outputs in reverse contact order. It never acts on the external reference. The declared accounts $H_C=\Delta(P_1+3P_2)$ and $H_S=\varepsilon I$ and all angular components are conserved. Physical formation, timing, action, recoil and service resources are separately specified for the selected deployment.

This gives **exact arbitrary joint carrier/reference delivery at 5τ** (four guide ticks plus receiver service) and **one packet per τ** for every finite funded horizon, with exact four-blank drain plus final receiver service. The selected implementation calibrates these timing counts and separately accounts for physical bandwidth, efficiency and capacity. For K admitted frames and four drain frames, the bill is **4(K+4) fresh tokens**, the matching raw-support and old-state outputs, four scalar drain blanks, complete controls/timing/decoder/terminal custody and separately supplied opening packets. Finite stock bounds finite operation; all support rails and coherences travel. Export is not local support catalysis or free disposal.

The uncorrected primary channel has $K_Z=Q+(1/3)L$, $K_x=-(2/3)|x><z|$ and $K_y=-(2/3)|y><z|$, where $L=|z><z|$ and $Q=I_C-L$. Its D_DIR covariance contrast is **8/9**. The exact transverse primary plane span{x,y} propagates under every finite composition; the primary longitudinal trace-distance error after four contacts is **6560/6561**, so joint recovery must not be called primary-only restoration. The primary symbol exp(−ika)M has **zero real curvature** and does not supply a complete transverse-only optical law. Own

calibration gives $v_g=a/\tau=v\le c$ *and* $n_g\ge 1$*; matching 1.468 would require independently established* $v/c=250/367\approx 0.6811989101$, admissibility rather than prediction, excluding endpoint service from the bulk comparison.

Lost/dephased side rails, correlated fresh tokens, missing old-state recipients, incomplete inverse, unaccounted opening/drain stock or blocked outlets defeat the relevant channel/account premises. This DEX4 embodiment has zero curvature and the stated primary transverse plane, rather than a complete transverse-only optical law. Exact quantity pricing is not physical formation or pricing; no free reset or indefinite closed source follows.

E11—prospective same-law boundary test N2. As an unselected DEX4 testbed, organize core, cladding and exterior lanes with nematic preparation contrast and retain the entire packet, side rails, longitudinal/quintet/scalar sectors, references and boundary outputs. The frozen full interface is $\mathbf{V_{total}=\otimes_r V_r}$. For L uniform lanes the primary spectrum has eigenvalues $\exp(-ika)$ and $\exp(-ika)/3$ with multiplicities **7L and L**. Exterior lanes share every transparent eigenvalue, so no isolating contour selects a core-only projector for that branch. Complete route leakage is exactly zero for every lane subset at every contrast, including zero: **disconnected routing, not boundary-generated confinement**. The full 729-state census has 81 scalar-, 243 triplet- and 405 quintet-carrier states; the reduced axial eigenvalues are not a spatial confinement theorem.

The axial segment control changes angular conversion and support exports without changing that transverse conclusion. The finite-span internal probabilities **1, 3/4, 1/2, 1/4, 0, 0** retain complementary exports; all old factors leave after four paid blank ticks, not through a derived reflecting endface. Full-packet inversion remains exact; dephasing/loss recoveries **11/27 and 1/81** are scoped rail-fault controls, not general link performance. The zero-contrast control rejects the posed confinement claim. A future candidate escaping this lane-class obstruction requires actual contrast-responsive transverse exchange/geometry breaking factorization; more lane labels, nematic axes, source masks or tariff labels do not suffice. A bulk operation changing this factorization is specified as a distinct transport law. The factorization limit applies to this lane class.

E12—prospective supplied-stock launch SCL1. Provide a full emitter E, finite pump seats A_n, scalar port blanks Q_n, a final scalar flush, retained source reference R, preselected readiness $b\in\{0,1\}$, and one charged scalar service cell/empty dump per attempted slot. All carrier seats use the same full nine-state space and equal declared tariff Δh. A full SWAP $A_n\leftrightarrow E$ brings in the supplied stock and exports the complete old emitter factor. Define $\mathbf{C_{EQ}=\Sigma_m(|s_E,t_{m,Q}\rangle\langle t_{m,E},s_Q|+h.c.)}$, zero on its other 75 states, and $\mathbf{V_b=\exp[-i\varphi(h_E+h_Q)]\exp[-ib\alpha C_{EQ}]}$, $\alpha=\pi/6$, $\varphi=\pi/8$. Each attempt performs the pump SWAP, V_b and compulsory service-to-dump SWAP in that order. Final full SWAP with the supplied flush exports the last emitter residue. No old bulk output funds the launch.

For ready attempts with independent supplied $\mathbf{\rho_r=(1-r)|s\rangle\langle s|+r\sigma}$, any triplet density σ and $\mathbf{1/25\le r\le 2/5}$, $\mathbf{p=r/4}$. Conditional triplet success preserves the entire angular/source-reference state; the unconditional channel retains failure, residue and purification. The exact per-attempt account is **pump $r\Delta$ + service Δ =**

launch rΔ/4 + residue 3rΔ/4 + dump Δ. Blank, unready, quintet, partial-transfer and unavailable attempts remain in the denominator; finite stock ends the run, not an unprinted retry. Boundary SWAP and F inject the same full angular port into each separately specified CRP2/RAR45 testbed. This is **not a common guided optical mode or an actual laser**; all bands/components and returned factors remain, with each bulk law's positive and adverse results intact.

The exact phase transfer is $\mathbf{G_{out}=\gamma_n{}^*\gamma_j G_{in}}$; independent phase-invariant stock gives no distinct-slot first-order coherence. Coherent stock is an input resource, not a generated mechanism. The unmodified same-cycle positive L0 realization is specifically excluded: $\boldsymbol{\alpha/\varphi=4/3}$, giving single-excitation minimum $\boldsymbol{-\Delta/3}$. This is a fixed-model sign obstruction, not universal pump impossibility. Missing reference/residue, unpriced service or blank stock, false triplet creation from blanks, outcome-conditioned readiness, a fitted phase/linewidth or a claimed guided mode without its same-law loader fails the corresponding gate.

E13—prospective scalar-fuel pump PFC1. Per attempt supply a scalar fuel F of tariff $2\Delta N_F$, two blank full seats A,B, a charged service/empty dump pair, readiness, retained fuel label and arbitrary spectator references. Fuel is number-diagonal with charged probability $\boldsymbol{2/25\le w\le 4/5}$, not a coherent source. Define the scalar pair $\boldsymbol{\Omega_{AB}=(|t_{+1},t_{-1}\rangle-|t_0,t_0\rangle+|t_{-1},t_{+1}\rangle)/\sqrt{3}}$, $u=|1_F,s_A,s_B\rangle$, $v=|0_F,\Omega_{AB}\rangle$, and $C_P=|u\rangle\langle v|+|v\rangle\langle u|$, zero on its full 160-state complement. Supply $\mathbf{H_P(b)=\Delta(2N_F+h_A+h_B)+b\hbar gC_P}$, $g=\Delta/(2\hbar)$, $gT=\pi/4$, and its complete exponential. Apply it, perform service SWAP, hand off A and retain B,F,labels/references. The coupling's physical existence is a premise, not a consequence of the intertwiner.

The ready first-attempt occupation is $\mathbf{r=w/2}$, with A triplet marginal $I_t/3$, and a **compulsory retained angular partner B at tariff rΔ**, occupation covariance $\mathbf{r(1-r)}$. No phase is presupposed. Repeated use of the same kit returns readiness to zero at $\boldsymbol{\ell=4}$; no irreversible recharge, bath or electrically maintained duty is supplied. Independent repetitions require new complete finite kits; the frozen run domain is $1\le M\le 16$. Omitted B, postselection on fuel conversion, false blank/unready signal or an unpriced fuel/blank/controller renewal fails the account. A repeat-operation selection supplies its own complete pump and renewal organization.

E14—prospective finite relative-phase tapping FRT1. Independently of PFC1 or any operating pump, supply one number-diagonal full reservoir R with triplet occupation $\boldsymbol{1/25\le r\le 2/5}$, $\boldsymbol{2\le K\le 16}$ scalar blank taps, service/dump pairs, and independent noise/retained-label bits with probability $\boldsymbol{0\le d\le 1/4}$. For slot j use the controlled phase flip D_j, identity for noise zero and $Z_R=\exp(-i\pi h_R)$ for noise one. Here τ_0 denotes the stipulated local update-cycle calibration $\Delta\tau_0/\hbar=\pi/8$. The selected clock implementation and physical time calibration supply this stipulated cadence. Supply the same full 75-complement-zero exchange C_{RAj}, $\theta=\pi/4$, $g=\Delta/(2\hbar)$, $T=4\tau_0$, and $\mathbf{U_{Hj}=\exp(-iTH_{0H}/\hbar)\exp(-i\theta C_{RAj})D_j}$, with $H_{0H}=\Delta(h_R+\Sigma h_{Aj})$. All taps, including emitted taps, receive the same free phase; no slot-dependent moving-frame repair is permitted. Apply service SWAP and retain the complete joint reservoir/tap/noise/reference state.

For j=1,…,K, $\mathbf{p_j=r/2^j}$ and, with $a_{jm}=|s\rangle\langle t_m|$ on tap j, the angular-summed correlation is $\mathbf{G(n,j)=\Sigma_m\langle a_{nm}\dagger a_{jm}\rangle=r\cdot 2^{-(n+j)/2}(1-2d)^{|n-j|}}$. This is within-block relative coherence generated by the declared shared contact, not a fitted G or an absolute mean field. $\mathbf{p_2-p_1=-r/4}$ exactly refutes stationary positive duty. This finite-block selection omits reservoir reset, repump and interblock phase locking; physical linewidth is calibrated separately. The independent same-marginal invariant-stock null and correlated-stock witness remain distinct. Wrong phase at the same populations, omitted noise labels/correlations, hidden replenishment or claimed indefinite finite-stock duty fails the relevant discriminator. A repeat-operation selection specifies reservoir reset, repump and interblock phase locking separately.

E15—prospective PFC1→FRT1→SCL1 composition. Hand the actual PFC1 A directly to FRT1 as R, retaining B,F and labels; no desired independent reservoir is inserted. After the phase contacts, pass each actual tap through the full unchanged SCL1 channel with its blank port, service payment and retained emitter residue, then the supplied final flush. The finite staged schedule is pump first, K phase contacts, then K separate launches, including storage until launch. Uniform single-excitation free phase or a common comparison frame is used, not slot-specific phase adjustment. The stated composition has **2K+4 fresh full-seat blanks**, **1+2K charged services and empty dumps**, finite fuel, noise/label/control inputs and all reference/outlet custody. Its selected implementation includes physical preparation, storage, controller, timing, action/recoil and reset resource accounts.

With r=w/2, for j=1,…,K the exact conditional consequences are $\mathbf{\ell_j=r/(4\cdot 2^j)}$, $\mathbf{G_Q=G_A/4}$, and $\mathbf{L_K=(w/8)(1-2^{-K})}$. At most one launch occurs per block; $\mathbf{P(\text{no launch})=1-L_K}$ includes unconverted fuel, blank input, reservoir residue and launch failure. Every slot and failed block remains in its declared denominator. For $z=2^{-K}$, the complete declared block balance is

$$\mathbf{[2w+1+2K]\Delta = [2(w-r)+r+rz+(3r/4)(1-z)+(r/4)(1-z)+1+2K]\Delta.}$$

The right-hand terms are respectively retained fuel, compulsory partner B, reservoir residue, source-launch residues, launched ports and service dumps. None is discarded on cancellation. When joined to a medium, add its initial state and separate old return factors; the internal launch cut cancels once, never twice. This is a tariff identity, not wall-plug efficiency, heat-per-bit or a physical source price. Independent blocks have no derived interblock coherence. Discarding B or launch residue, replacing the shared state by a favorable marginal, using a fitted G, or omitting failure/service/flush stock defeats the composition. The unchanged SCL1 $\mathbf{-\Delta/3}$ same-cycle obstruction survives; the pump/phase contact duration does not repair it. This correlated source does not prepare the full-future-product input specified for E16.

E16—prospective collection and record law OCCR1. Each finite attempted arrival has a distinct one-shot capsule: full detector D=s, charged scalar bias B and empty terminal T, orthogonal degenerate scalar pointer states u,v with initial u, charged service S and empty dump Z, preselected readiness b, and an external full angular port Q. All source/bulk/reference factors and returned port states are retained. On

Q,D use the full exchange C_{QD}, zero on its other 75 states, and $\mathbf{V_b=\exp[-i\varphi(h_Q+h_D)]\exp[-ib\alpha C_{QD}]}$, $\alpha=\pi/6$, $\varphi=\pi/8$. With $X_P=|u\rangle\langle v|+|v\rangle\langle u|$ and full scalar SWAP W_{BT}, the complete conditional write is $\mathbf{L=(I_D-P_{t,D})\otimes I_{BTP}+P_{t,D}\otimes(W_{BT}\otimes X_P)}$. The complete encounter is $\mathbf{A_b=W_{SZ}LV_b}$, rightmost first. No projection deletes failures. Occupied/nonfresh capsules retain this same full unitary but do not inherit the fresh-blank one-way claim.

For either exact full angular testbed port, $\mathbf{p=b\cdot tr(P_t\rho_{arrival})/4}$. Triplet success preserves the full angular/reference state in retained D; collection is the terminal/pointer predicate T=1,P=v. Failures, blanks, quintets and unready attempts stay in the denominator. Let r and q here denote arrival triplet and quintet probabilities and $\mathbf{x=\Delta(r+3q)}$. The closed per-attempt declared account is

$\mathbf{x+2\Delta = 2p\Delta}$ **collection +** $\mathbf{(x-p\Delta)}$ **residue +** $\mathbf{\Delta}$ **dump +** $\mathbf{(1-p)\Delta}$ **unused bias.**

The collected $2p\Delta$ consists of two different excitations: arrival in D and supplied bias in T. Unused bias is retained, not redeemed; service is paid on every attempt. No free reset or medium-depletion funding occurs. CRP2's 16-stipulated-update transfer retains the optical constraints of E7. RAR45's exact arrival functional includes slot-8 weight **1/256**, collection **1/1024**, and the loaded-reference, reuse and drain limits of E8. These are separate unselected full-port testbeds, not an N2 guided mode or detector identification; no DEX4 collector follows by naming its receiver.

The one-shot pointer is retained and exactly distinguishable under stipulated isolation. **Full stability uses separately supplied T_product**, a full-future-product preparation requirement. In broader T_full traffic, decorrelation is excluded by exact invariant-sector witnesses: **CRP2 pointer/future covariance 1/16**, **RAR45 slot-8/slot-10 covariance −49/1048576**, with record/bulk covariance **−93/262144**. Isolation does not delete existing future correlations; neither unchanged marginals, no-signalling nor fresh capsules proves decorrelation. The unmodified same-cycle positive D0 route has $\mathbf{\alpha/\varphi=4/3}$, minimum $\mathbf{-\Delta/3}$, and is specifically excluded, not every physical-memory mechanism.

Missing or altered full-reference transfer, omitted bias/service/residue, a claimed reset without recipients, or retained future-accessible covariance contrary to the frozen product target fails the corresponding claim. The selected record embodiment specifies traffic preparation, collection, storage, clock/energy implementation and a compatible port.

Common boundary of E7–E16. Each construction or scoped obstruction applies to its printed state, preparation, contact law, timing, controls and recipients. Physical clock, constituent, carrier, source and detector implementations are selected for that domain and included in the accounts. These embodiments do not alter E1's mathematical version or transfer performance, capacity, bit-rate or efficiency between different laws.

E17—prospective exchange-record representation. The representation applies the joint state/update/readout discipline to the complete declared account. In a two-opening example, the source, openings, receiver and retained distinguishing records belong to the same arrangement; a whole registration and its accumulated fringe pattern are distinct descriptions. No universal clock or unseen trajectory is inferred. A phase-sensitive example includes equal diagonal populations with later counts

1/10 versus 9/10: future phase-sensitive distinguishability, not today's marginal alone, governs reduction. Persistent site-labelled clock memory coupled to recombining order histories diagnoses the specified CT1 computational burden, not universal undecidability or a physical no-go. (See E17, paragraphs O2-014–O2-016.)

[O2-013] **Mathematical framework.** Whole-turn action, co-signed exchange, relative turning and retained angular accounts provide the POAMS interpretation under test. Finite complex states, reversible operators, tensor composition and trace/Born readout are the specified Coherent Completion Kernel (CCK) mathematical bridge. CT1 contact tariffs and CF1 instrument couplings are candidate-specific conditional inputs. Trace and diamond distances and bounded-sample concentration are imported mathematical tools. The score map, schedule, bins and custody fields are disclosed operational definitions. (See E18.2–E18.7.)

[O2-014] **Joint state and record.** A state includes the source preparation and every retained reference needed by the future word. An update is an ordered operation on that state, including controls and archival handoff. A readout is the specified probability or mean of retained output records after that word. Off-diagonal blocks remain when a requested record is phase sensitive. A partial trace in a calculation is not a physical erasure, reset, fresh preparation or disposal of an apparatus recipient. (See E18.1 and E18.8.)

[O2-015] **Persistent-clock CT1 representation.** For the specified CT1 cycles the bulk evolution has the exact form

$$U^t = \Sigma_q |q+t\rangle\langle q| \otimes V_q(t).$$

Here q denotes the collection of initial three-label site clocks, the additions use the clock's cyclic labels, and V_q(t) retains the original order of interactions. Each site has a three-component record interface and a Gram kernel of rank at most three. Its initial clock label persists through revisits. This contracts histories mathematically without refreshing clocks. The identity belongs to CT1's complete-cycle word; the RCM-1 word below has no intervening clock advance. (See E18.2.)

[O2-016] **Permitted reductions.** A retained distinction may be merged only if every admitted future operation and requested record has the same probabilities in the merged descriptions. A record may be summed out after its last relevant contact or following an equivalent future-effect sufficiency proof. No general efficient procedure for producing that proof is asserted. The nominal magnetic-deficit-at-most-two edge factorization does not cover arbitrary old/fault stock. The source-count reduction uses six positive population contractions, not a replacement by a pure bright input; two have nonreacting dynamics and four still require coupled calculation. Coherence averages amplitudes; population averages full populations. Squaring the former generally does not give the latter. (See E18.1 and E18.3.)

E18—prospective RCM-1 same-bank repeated-contact diagnostic and apparatus. This embodiment interrogates a contact-memory ingredient separately from useful optical transfer and the E1 payload channel. Two contacts are not two CT1 cycles. CT1 retains its separate t=10 tests $|g_-| \geq 4e{-}6$ and $p \geq$

1e−8. The RCM-1 apparatus is configured with the complete preparation, ordered update, record, certificates, controls and decision procedure below, including AP-RCM and AP-M as applicable.

E18.1 — Full inventory and nominal preparation

[O2-017] Retain the complete finite 65^3 inventory, both C and D carrier rails, order organizations A, Bs and F, the hard-Q nominal code, outlets, references, flags, command/history stock, old outputs and their physical complements. Use the original 125-site source bank and its normalized Fourier profile. The selected address is d=+1, center −8 e_z, central mode q=+e_z. The receiving selector is addressed to the same bank and matched mode with distinct fresh receiver stock. This selects a diagnostic address, not a successful CT1 transport direction. (See E18.2–E18.3.)

[O2-018] Nominal bulk orders are u_plus=(x+i y)/sqrt(2), scalar clocks are $\chi=(|0\rangle+|1\rangle+|2\rangle)/\sqrt{3}$, and carriers/outlets begin in vacuum under the supplied preparation. The source input for the decisive phase history is (vac+u_minus)/sqrt(2), where u_minus=(x−i y)/sqrt(2); its phase is relative to the retained reference. Keep the complete original source selector and its old reference, flag and longitudinal reactions. The input specification is not a population premeasurement or substitution of a pure bright source. Population histories, if separately considered, retain their actual mixed source and positive receiver effect. (See E18.3–E18.4.)

E18.2 — Clock-conditioned contact word and AP-RCM

[O2-019] At each site the existing conditional operators are

```
G_C = Σ_q |q⟩⟨q| exp[−i θ_q (S_CA−I)],
G_D = Σ_q |q⟩⟨q| exp[−i θ_(q+1) (S_DBs−I)],
θ = (π/4, π/6, π/8).
```

The q in this local expression is the scalar clock label 0, 1 or 2, with q+1 taken cyclically; it is distinct from the bank-mode label. S is the carrier/order swap on the V1⊗V1 block. The indicated generator is zero on carrier vacuum and quintet complements. Each admitted inverse is the full adjoint, including complement action. The scalar clock is not an unbooked angular reservoir. (See E18.1 and E18.4.)

[O2-020] Define F=product_sites G_D G_C and W_P=F F. The same physical clock appears in both encounters. There is no intermediate transport, X_T clock advance, order permutation, edge matching, Q change, mixer, singlet or outlet operation. Unselected operations are identity in this separate program; none is deleted from a claimed CT1 cycle. Identity spectators remain physically present and accounted. Nominal reduced blocks may simplify prediction, but arbitrary old/fault input is governed by the full operators. (See E18.4.)

[O2-021] **AP-RCM configuration.** The apparatus is configured to provide selective existing-contact programming and same-bank receiver addressing with distinct receiver stock. It preserves both rails, all 125 bank sites, the 65^3 inventory, original source/receiver laws, θ/χ, inverses and complements. The selected itinerary excludes extra contacts, clock advance, altered tariffs, nonseparable routing, reference reuse and changed source laws.

E18.3 — Phase-sensitive receiver and complete records

[O2-022] The CF1 joint carrier/flag transition uses |g⟩=|0,0_f⟩, |e_a⟩=|1,a;1_f⟩ for a=x,y,z, L_a=|g⟩⟨e_a|, E=Σ_a |e_a⟩⟨e_a| and G=|g⟩⟨g|. It acts on (V0⊕V1⊕V2)⊗C2_f. The transition changes carrier and flag together; it is not a flag eraser. The actual reacting phase stock R1p has the like carrier representation. R0p, selector references, all flag branches and inactive/old stock remain retained. (See E18.3, paragraph O2-023.)

[O2-023] At each receiving rail/mode, apply relative phase to R1p and then Uj=exp[−iπ(B+B†)/4], with B=Σ_a L_a†⊗d_R1p,a and d_R1p,a the phase-stock lowering transition. Record actual D_dark into the f=0 phase pointer, then D into f=1, both after Uj. With N_R=P1 on the phase stock,

```
D = E⊗I − I⊗N_R,
Pact = E⊗P_R1p,0 + G⊗N_R,
D_dark = D(I−Pact),
Urec(O) = Σ_m Π_m(O)⊗X_pointer^m,  O = D_dark or D.
```

The spectra are −1,0,1 and the pointer is a full C9 cyclic register. The commuting output observables permit the specified ordered records. Nonblank pointers retain the same cyclic action; they are not silently reset or unwrapped. Population pointers are separate and inactive in these phase histories. (See E18.5.)

[O2-024] The balanced readout is A=(L_C+L_D)/sqrt(2), not one rail or the arithmetic average of the rails. Actual DARK can be nonzero even for nominal no-quintet input. Its measured record is kept and used in the specified arithmetic; it is never replaced by a constant, a newly prepared dark sample or an event-selection rule. The 500 receiving phase-pointer outputs — 125 modes, two rails, two f slots — are retained as raw labels after protected archival handoff. Only the central matched four contribute to the decisive score. Source/inactive/population pointer factors remain retained unread, not assigned fabricated labels. (See E18.5 and E18.9.)

E18.4 — Persistent and independent-renewal predictions

[O2-025] In the nominal nonvacuum branch the contact-only word preserves carrier site and rail. For one occupied site the relevant opposite-order pair is e0=|u_minus⟩_C|u_plus⟩_A and e1=|u_plus⟩_C|u_minus⟩_A, or its D/Bs counterpart. Because S^2=I, the returned-order coefficient after two same-clock contacts is

(1+exp(4iθ_q))/2. Averaging these coherent coefficients over the retained χ gives

$$A_P = 1/4 + i(2+\sqrt{3})/12.$$

The D offset permutes the same three terms. The coefficient is the same at each occupied seat, and the matched normalized spatial overlap is exactly 1. This is a full-size nominal invariant calculation, not a reduced physical specimen or a simulation of arbitrary fault stock. (See E18.1–E18.2.)

[O2-026] The full-source/receiver coherence contractions give M_mem=A_P/4 and C_mem=A_P/8. Consequently the persistent quadratures are Δ_(π/2),P=1/32 and Δ_0,P=−(2+√3)/96. These are unconditioned signed phase-record means. They are not probabilities of successful transfer, and useful population is not $|A_P|^2$ or $|M_{mem}|^2$. (See E18.3 and E18.6.)

[O2-027] In the independent-renewal alternative H_R, every site supplies an independent χ clock for the second encounter and retains the first clock and its correlations in export stock. The order is not renewed. The first-contact reduced behavior agrees with H_P. Its two-contact quadrature is

$$\beta = (15-2\sqrt{3}-\sqrt{2}-\sqrt{6})/288,$$
$$25/960 < \beta < 26/960.$$

The persistent prediction is exactly 1/32, and the separation is more than 1/240. The renewal alternative is a separately accounted hypothesis, not an approximation licensed for persistent CT1 evolution. (See E18.6–E18.7.)

E18.5 — Frozen total score and full-domain certificate

[O2-028] Freeze the score s=[0,1,0,0,0,0,0,0,−1] over raw labels 0…8. In pointer order (T_C,D_C,T_D,D_D)=(TOTAL_C,DARK_C,TOTAL_D,DARK_D), define for each scheduled trial i and row j:

$$k_i = s(T_C)-s(D_C)+s(T_D)-s(D_D),$$
$$Y_i = k_i/\mathrm{sqrt}(2),$$
$$r_j = (\Sigma_i\, k_i)/(n\ \mathrm{sqrt}(2)).$$

Each k_i is an integer in [−4,4]. The score is total over all 6,561 four-pointer tuples. Labels 2–7 retain their distinct raw identities although their arithmetic scores are zero. Unreadable or missing labels are not zeros. Trials have weights 1/n with no successful-event denominator, clipping, blank reinterpretation, rail averaging or DARK replacement. Within-trial correlations are allowed. (See E18.3 and E18.9.)

[O2-029] The full-domain observable norm is $\mathbf{2\sqrt{2}}$, with range $\mathbf{4\sqrt{2}}$. Certify, independently of the discriminator outcomes,

$$4\sqrt{2}\cdot D_{prep} + 2\sqrt{2}\cdot \delta_{word} + e_{pointer} + e_{stat} \le 1/960.$$

D_prep is the half trace-norm distance of the complete jointly supplied state from the declared nominal state, including relevant external references and all stock. δ_word is the unhalved diamond distance of the complete ordered channel through protected pointer retention, including full fault/leakage/complement action. e_pointer bounds physical decoder/calibration bias outside that channel cut. e_stat is the fixed-sample statistical term. The norm-2 certificate is not applicable to this two-rail full-domain score. (See E18.5, paragraphs O2-030–O2-031.)

[O2-030] The default ceilings are $D \le \sqrt{2}/30720$, $\delta \le \sqrt{2}/15360$, $e_{pointer} \le 1/3840$ and $e_{stat}=1/3840$. Their four boundary contributions are each 1/3840. Bounds must come from disjoint calibration or independently established structural evidence, with explicit domains, conversions, uncertainty/failure probability, drift coverage, validity interval, before/after calibration evidence. Marginal fidelities, a blank-pointer count or a label saying “fresh” do not establish the required joint bounds or independence. (See E18.7.)

[O2-031] A sufficient joint four-label total-variation decoder bound η gives $e_{pointer} \le 4\sqrt{2\eta}$; independently valid per-pointer mismatch bounds η_a give $e_{pointer} \le \sqrt{2\Sigma_a \eta_a}$. Every label and raw decoder output remains retained. These are mathematical conversion bounds, not observed decoder performance. The exact integer $K=\Sigma_i k_i$ and $r=K/(n\sqrt{2})$ permit rational-pair arithmetic; decisions use exact algebra or certified outward intervals, not a floating-point threshold guess. An interval enclosing r is not expanded by the already budgeted total error a second time. (See E18.7 and E18.9.)

E18.6 — Ten-row protocol and mimic/sham configurations

[O2-032] Freeze $a=1/32$, $b=\beta=(15-2\sqrt{3}-\sqrt{2}-\sqrt{6})/288$, $c=(2+\sqrt{3})/96$, $d=(1+\sqrt{2})(2+\sqrt{3}+\sqrt{2})/288$ and $t=(7+\sqrt{2})/96$. Here d and t are table constants, not the source address or CT1 cycle count. Use the following complete schedule. All rows use the same full inventory and actual four-pointer score, except the expressly stated input, phase or word change. (See E18.4.)

ID	Scheduled posing	H_P ideal	H_R ideal
N	Two contacts; u_minus; $\varphi=\pi/2$	a	b
Q	Same two contacts; $\varphi=0$	−c	−d
V	Vacuum source; two contacts; $\varphi=\pi/2$	0	0
I	Identity/no contacts; same matched bank; $\varphi=\pi/2$	1/8	1/8
S	u_plus input/reference; two contacts; $\varphi=\pi/2$	1/8	1/8
O	One contact; u_minus; $\varphi=\pi/2$	t	t
R	N with negative relative source coherence	−a	−b
M	Fresh-clock SWAP/export between contacts	b	b
H	Added swap slots; SWAP then SWAP back before second contact	a	b
Z	Incoherent half-vacuum source mixture; preparation records retained	0	0

[O2-033] **AP-M configuration.** For M and H, the apparatus is further configured to provide scalar-clock swapping/export and independent fresh χ stock, without order renewal, hidden return interaction or unintended phase change. The AP-M configuration is additional to AP-RCM. For M, use a full C3 SWAP at every site: the first clock and its correlations are exported, and independent χ enters the second contact. For H, use the same fresh stock and two swaps per site, with no intervening contact, so the added pair composes to identity on arbitrary inputs. H_R still independently renews at its second encounter. Z may use an independent equiprobable source-sign bit, retaining both classes and the bit without selecting a class. (See E18.2 and E18.8.)

[O2-034] Let N_sites=274625. M adds N_sites fresh C3 factors and N_sites swap letters; H adds the same number of factors and 2N_sites swap letters. Added recipients remain protected outside the baseline data/archive modules, with no reset, reuse or tracing credit. The finite program is enlarged by those letters. Physical routing, extraction and implementation-specific apparatus require additional independently charged accounts. Failure to implement or certify any required control makes the scheduled comparison inconclusive. (See E18.8.)

[O2-035] Use n blocks, each containing the table's ten IDs in order cyclically rotated by (block−1) mod 10. Seal IDs, addresses, stocks, times, settings and word before observation. No outcome-driven grouping, substituted trial or favorable retry is allowed. A safety or receipt abort is logged; an incomplete fixed schedule is inconclusive. Deterministic balancing does not prove independence or remove drift. (See E18.7 and E18.9.)

E18.7 — Statistics and decision procedure

[O2-036] With e_stat=1/3840, the frozen sufficient sample count is $n=256\cdot 3840^2$=**3,774,873,600 trials per row**, with **37,748,736,000** nominal slots across ten rows. The selected acquisition stock is sized for this finite sample schedule. For independently qualified bounded trials, which need not be identically distributed, the row mean obeys

$$\Pr(|r-\bar{m}| > e_{stat}) \le 2\exp(-n\, e_{stat}^2/16) = 2\exp(-16).$$

The ten-row family bound is $\le 20\exp(-16) < 20/2^{16} = 5/16384 < 1/1000$. Freeze $\alpha_{stat}=1/1000$. The statistical interval $[r-1/3840, r+1/3840]$ feeds the budget; it does not further enlarge the already error-expanded bins. Unqualified cross-trial correlations invalidate this plan. No optional stopping, fitted variance or post-outcome effective-sample-size repair applies. (See E18.5.)

[O2-037] Preparation/word/decoder certificates are deterministic valid bounds or jointly have $\alpha_{cert} \le 1/1000$ from disjoint calibration. The combined conditional comparison error probability is then $\le 1/500$, or $\le 1/1000$ for deterministic certificates. The absolute mean-error bound holds on that simultaneous good event, not on every random sample without exception. These probabilities bound the qualified inference procedure with the specified AP-RCM configuration. (See E18.5 and E18.6.)

[O2-038] Use inclusive bins B_P=[29/960,31/960] and B_R=[24/960,27/960]. For an admissible outward arithmetic interval I_N enclosing the primary r_N:

1. If any record, certificate, schedule, custody, AP-RCM/AP-M or statistics requirement fails, report **INCONCLUSIVE/INADMISSIBLE**, never passage.
2. If I_N is wholly contained in B_P, report the primary label **PERSISTENT-COMPATIBLE**.
3. If I_N is wholly contained in B_R, report **RENEWAL-NOT-EXCLUDED**.
4. If I_N is disjoint from both bins, report **REJECT-BOTH-FROZEN-ENVELOPES**, or failure of posing/error conditions.
5. Otherwise report **INCONCLUSIVE**, including intervals spanning bins or touching a bin without containment.

A singleton endpoint belongs to its inclusive bin. The shortcut r>28/960 is not this rule. (See P5–P6.)

[O2-039] Evaluate common controls V,I,S,O,M,Z against their exact targets ±1/960. Entire outward intervals must be contained for compatibility; disjointness is discrepancy and partial overlap is inconclusive. A common-control discrepancy prevents primary-bin passage: preserve the arithmetic label and report the complete comparison INCONCLUSIVE/CONTROL-DISCREPANCY. Q,R,H have separate H_P and H_R target bands. Test each hypothesis against its own bands; do not demand persistent predictions as a prerequisite for testing renewal. If independently qualified conditions and common controls hold but that hypothesis's specific control interval is disjoint, its full signature is rejected. If N already excludes the other model, report both frozen signatures rejected, or an unrecognized posing/error failure. No unique replacement theory follows. Controls may not rescale N or fit phase, contact angle, decoder, DARK baseline, error allowance or bins. (See E18.6 and P5–P6.)

E18.8 — Archive, controller and account closure

[O2-040] A finite reversible program may use $|j\rangle\psi \to |j+1 \bmod L\rangle V_j\psi$ for its listed letters. A matched independent archive receives the complete data and history by full SWAP, carrying correlations rather than erasing them. The program counter P and active/off history remain separate. The actual final externally commanded OFF-token SWAP lies outside the A-controlled unitary program; moving an A-changing SWAP inside that controlled construction would not supply the specified unitary. Protect the archived result from later admitted contacts before extracting pointer labels. (See E18.9.)

[O2-041] The unchanged sufficient baseline one-shot account includes 1,098,500 contact letters, 1,134,997 apparatus/contact/history letters, 72,230,874 archive letters, L=73,365,871, 73,365,872 strokes including OFF, and 146,084,149 listed factors before idle CT1 stock. Whole-Q exchange charges its full recipient inventory rather than equating one listed stroke with one microscopic recipient. Mimic/sham stocks and their enlarged programs are additional. These finite listed-primitive accounts are separate from

microscopic pricing, dissipation and capacity calibration, and are not funded by the CT1 service bill. This one-shot configuration omits autonomous repeat, stock regeneration and indefinitely reusable service. (See E18.6.)

E18.9 — Record and custody implementation

[O2-042] Maintain the following linked records with immutable raw originals and a separately identified derived tree. Numeric metadata uses integers, numerator/denominator strings or exact algebraic coefficients; nonfinite values are not admissible. Binary assets have immutable location, byte size and SHA-256. (See E18.8.)

- **Run manifest:** protocol/source/code versions and hashes; operator; specimen and full site/rail/mode inventory; geometry, wiring, orientation, environment and calibrated timing; source address; ordered word and complete schedule; stock and recipient IDs; decoder and independent calibration IDs; all four budget terms and confidence certificates; fixed n and stopping rule; acquisition authorization identifier; retention capacity and replication plan.
- **Stock/account manifest:** every source, reference, flag, pointer, phase, clock/order/Q, outlet, command/history, archive and OFF factor; fresh versus retained status and preparation evidence; swap/export parent and recipient links; inverses and complements; final protection evidence; M/H additional factors and strokes.
- **Trial schedule record:** unique run/block/row/trial ID, exact settings, word, address, stocks, expected timing and budget-coverage references; cancelled or unstarted slots remain tombstones, never reused IDs.
- **Raw trial receipt:** commanded and observed settings; raw instrument bytes and units; every calibration transform; contact/phase/record/archive/OFF timing; all 500 receiving phase-pointer labels or explicit missing/unreadable status; central four links; retained-unread links for other pointer factors; initial blank certificate; alarms, dropouts and all old outputs. No carrier/clock premeasurement is introduced to manufacture hidden-path metadata.
- **Reduction receipt:** raw hashes, frozen decoder version, nine-label marginals, all 6,561 joint four-pointer cells, trial labels/k_i/Y_i, expected/present/missing counts, exact K and r, arithmetic and statistical intervals, separate systematic/sampling terms, every control result, primary label and full disposition.
- **Custody chain:** append-only chunk ordinal, parent hash, bytes, SHA-256, creator/time, complete ordered manifest and seal; independent rehash/reduction log and replay plan. Seal protocol, schedule, decoder and calibration before the first trial; chunks on close; complete raw manifest before analysis. Keep all aborted receipts and scheduled tombstones.

[O2-043] Cross-record checks require exactly the sealed schedule, unique IDs, no unlisted substitutions, all 500 active receiving pointer observations for each completed trial, and retained-unread custody links for every other pointer factor. The central order must resolve to

TOTAL_C,DARK_C,TOTAL_D,DARK_D. Raw assets and certificate links must rehash. Histogram marginals must agree, the joint histogram must sum to the full scheduled n only when complete, and its score sum must equal K. Missing slots invalidate the full comparison. Schema validity is not evidence of source state, independence, time, certification or physical provenance. (See E18.5–E18.7.)

E18.10 — Independent failure conditions and scope

[O2-044] A qualified interval in B_R excludes the persistent envelope subject to renewal controls; a qualified interval in B_P excludes the accounted renewal envelope subject to persistent controls. An admissible interval disjoint from both excludes both frozen envelopes or reveals posing/error failure. Disjoint Q, reversal or sham signature bands reject the corresponding signature; discrepant vacuum/dephased, identity/saturated, one-contact or mimic controls prevent passage. M must meet its exact target band; coarse membership in B_R alone is insufficient. No adverse record may be fitted away by choosing a new source distribution or clock tariff. (See E18.6–E18.7 and P5–P6.)

[O2-045] The method comprises establishing the existing contacts at fixed θ; selecting identity spectators, same-clock retention and separable same-bank routing with distinct stocks; preparing the full inventory and source; applying the CF1 read order and four-pointer reduction; establishing global preparation/channel bounds; fixing the decoder and qualifying the statistics; protecting the complete archive and OFF record; supplying AP-M stocks and exports; and maintaining conservation accounts. The selected embodiment excludes unbooked angular momentum, altered Q/tariffs, missing recipients, unintended recontact, hidden renewal and order replacement.

[O2-046] RCM-1 provides the nominal mathematical discriminator and fixed acquisition protocol specified in E18.1–E18.9. Material formation, apparatus production, calibration and repeat-operation arrangements may be selected per deployment with their own accounts. The comparison addresses the defined persistent-contact memory ingredient; CT1 optical-transfer thresholds and the E1 communications channel retain their separate operating domains.

Common boundary of E17–E18. E17 specifies exchange-record representation; E18 specifies the RCM-1 diagnostic configuration with AP-RCM and the additional AP-M controls. Each retains its stated inventory, preparation, clock word, output records and accounts. E1–E16 keep their separately defined contact laws, technical limits and operating domains.

PROPHETIC EXAMPLES

P1—opposite source symbols. For a zero, the source executes +Z; for a one, it executes −Z. With fixed Z receiver, the ideal outcomes are respectively completed plus and completed minus. The receiver maps only those physical pointer codes to zero and one. Replacing the actual pointer while retaining the source label must replace the decoded value or yield erasure; retaining the pointer while changing hidden source metadata must leave the decoded value unchanged.

P2—severed guide. The literal CUT operation prevents delivery across the source edge while preserving the scheduled wrapper and phase operations. No bypass leaf path exists in the tree. A decoder that still recovers withheld symbols through source labels or correlated traffic would fail the payload-separation gate. The control retains complete records for evaluating leakage.

P3—retained work at a changed tariff. For a completed local transfer from κ to $7\kappa/8$, the terminal receives $7\kappa/8$ and the net work account receives $\kappa/8$. Both are retained, and the same source factor continues through the endpoint and return operations. A terminal-only reading of $7\kappa/8$ is not the whole delivered-energy result.

P4—bounded finite message. A predetermined number of attempted symbols is supplied with the corresponding complete fresh stock. The recorded output contains all symbols and erasures; it is not shortened after failures. Native posts, material stock and positive capacity scale with the actual allocated units. No numerical SI throughput, link distance or electrical input/output performance is assigned before independent calibration and physical interface realization.

P5—prophetic persistent-contact comparison. With the E18 configuration, required controls, complete records and independently established certificates, an admissible N interval wholly within [29/960,31/960] with the persistent model's Q/R/H controls compatible would be persistent-compatible and exclude the stated renewal envelope at the protocol's conditional inference level. It would test the contact-memory ingredient, not establish a working fiber or the POAMS ontology.

P6—prophetic renewal or failed comparison. Under the same independent qualification, an N interval wholly within [24/960,27/960] with the renewal model's controls compatible would exclude the persistent envelope while leaving renewal not excluded. A qualified interval disjoint from both bins would reject both frozen envelopes or the posing/error conditions. Missing controls, unmet configuration requirements, incomplete slots, unqualified independence or intervals not wholly classifiable would be inconclusive, never passage. No record is rescaled to manufacture a preferred bin.

POAMS-NATIVE DIRECT-OPERATION EMBODIMENT

The native physical claim concerns circulation, angular momentum and co-signed exchanges within one connected organization. Work is not a second invisible supply: its scalar circulation states and positive activation stores are retained. The finite spin spaces, projectors, unitary/Born calculus and concentration bounds are openly imported mathematical representations. Common-coordinate matching, positive contact types, formed correlated bodies, native cadence, thermal opportunities and atomic capture are explicit constitutive premises.

The physical implementation selects guarded permutations and controlled spectral operations, with material fabrication, an external symbol-setting interface and occurrence-to-digital readout configured for the specified contact laws. The complete source program and engineering construction are specified for those operations.

Pair-posting selection. The admitted-event pair posting $\delta=-2(p\cdot\xi)\xi/\xi^2$ is conditional on fixed nonzero spacelike endpoint separation, fixed endpoint norms, no extra retained recipient, total energy–momentum and the complete six-component event moment. A distinct posting requires $(p+q)\cdot\xi=0$ with $p\cdot\xi\neq0$; identity remains possible and the matched repeated pair operation is an involution. The theorem does not select admitted graph, readiness or succession. Pair admission and bulk transport are separate constitutive selections at two composition levels. CRP2 and the E8–E16 operations retain their own separately specified laws.

Material and calibration selections. Each deployment selects its material organization, input/output interfaces and work, length and time calibrations for the specified contact law. Optical quasiphase and a conserved tariff remain distinct from an $E=h\nu$ energy identification. Attenuation is funded by the source, and all complements, references, unsuccessful operating outcomes and ordinary outlets remain in the complete account.

DISCLOSURE REFERENCES AND RELATED PROVISIONALS

Subject	Reference within this disclosure
Source-to-output architecture, tree guide, contacts and receiver	FIGS. 1–5; E1–E4
Finite operation, resources and contact compilation	FIGS. 6–10; E5–E6
Transport, source and record constructions	E7–E16
Exchange-record representation and persistent-clock reductions	E17
RCM-1 inventory and AP-RCM contact configuration	E18.1–E18.2
CF1 joint phase instrument and nominal predictions	E18.3–E18.4
Full-domain score, certificates, AP-M controls and decision procedure	E18.5–E18.7
Archive, accounts, custody and operating conditions	E18.8–E18.10
Illustrative operating cases	P1–P6

Related portfolio references are **PROV-C, Instruments & Metrology**; **PROV-E, Media & Arrays**; **PROV-F, Memory & Signaling**; **PROV-G, Time-Rate Management**; **PROV-H, Computation**; **PROV-M, Addressed Adjacency**; and **PROV-N, Counterfactual Receipts**. These title-level references do not substitute for the RCM-1 teaching, score, bins or configuration requirements set out in this disclosure. No priority claim or incorporation by reference is made by this list.

CLAIMS

1. A fiber communications system comprising: a source assembly retaining an angular signal organization and a reference organization; a source-local selector configured to encode a user symbol by a source angular operation after a common readiness condition; an elongated guide comprising a tree of signal sites and local contacts; scalar work pairs associated with the contacts and configured to exchange complementary work states during transfer between signal sites; a

receiver configured to apply a fixed angular analysis; a receiver-occurrence decoder configured to map a completed receiver pointer to a symbol or erasure without consulting source-setting metadata; and a controller configured to retain work and unsuccessful-attempt records through a finite operating sequence.

2. The system of claim 1, wherein the selector chooses opposite angular commands while the receiver analysis basis, guide update word, endpoint settings and launch horizon remain unchanged.
3. The system of claim 1, wherein each signal site is represented by a complete singlet, triplet and quintet organization and the local contact preserves specified behavior on wrong-work, noncode and additional-occupation states.
4. The system of claim 1, wherein a support supplies a common stationary coordinate setting a signal tariff and an owned work-dimer tariff at a ratio of two to one, and a support-load operator commutes with the specified guide-edge exchange, triplet-tag, guard and presence operations.
5. The system of claim 1, wherein both complete work pairs for a root-oriented contact are mounted on its arrival side and a completed crossing transfers a signal tariff difference into the retained work pairs.
6. The system of claim 1, wherein a unique root-to-vertex path determines a retained work configuration and reverse crossings restore the traversed work words without erasing a path record.
7. The system of claim 1, wherein the guide comprises a backbone and nonzero transverse leaf contacts without leaf-to-leaf axial connections, and a fixed loader, update word and translated catcher act independently of the selected user symbol.
8. The system of claim 1, wherein the decoder assigns zero and one only to respective completed plus and minus pointer codes and assigns incomplete, rejected, noncode or unavailable pointer codes to erasure.
9. The system of claim 1, further comprising a separate audit processor permitted to inspect source settings and causal completion records, wherein audit access to those fields does not provide or repair the payload value assigned by the receiver-occurrence decoder.
10. The system of claim 1, further comprising a full-state equal-tariff exchange configured to move the actual received source and its retained correlations into a local endpoint, a driven relative-phase operation at the endpoint, and a priced source-return route retaining spent work, measurement and thermal states.
11. A method of communicating a finite user message, comprising allocating complete physical stock for a finite number of attempted symbols; preparing a source and retained reference at one initial qualification boundary; selecting each symbol through a source-local angular operation; conveying the angular organization through a work-compensated tree guide; applying fixed receiver analysis; decoding actual receiver pointer occurrences independently of source-setting metadata; retaining erasures and failures in the attempted-symbol record; and supplying a fresh complete unit for a subsequent attempt unless a separately qualified reset or replenishment operation is performed.

12. The method of claim 11, further comprising maintaining the same padded source-command and launch horizon for opposite symbol choices to exclude an earlier command-dependent timing message.
13. The method of claim 11, further comprising completing a local phase comparison and return of the same source, wherein the local endpoint phase setting is not used as the remote source payload control.
14. The method of claim 11, further comprising maintaining separate bounds for symbol error, erasure, complete-record total variation and retained-state distance, and recalculating an affected bound when an operation, interface or schedule changes.
15. The method of claim 11, further comprising allocating independent complete units or establishing a per-history conditional record law before applying a repetition-code or concentration bound.
16. A work-compensated guide contact comprising two complete signal dimers, two scalar work pairs with complementary ready and spent words, and a positive guarded interaction configured to exchange the triplet signal and the work words on an admitted active sector while retaining the specified action on its full complement, the contact being arranged under common-coordinate tariff matching to conserve complete local energy and total angular momentum.
17. The contact of claim 16, wherein the positive guarded interaction has generator $K=(PL+PR-T-T\dagger)/2$ on the admitted opposite-direction active columns and is zero on their orthogonal complement.
18. The contact of claim 16, wherein the interaction is accompanied by two positive endpoint phase operations to obtain a phase-flat guide gate, rather than treating the contact's active-block phase as a global phase.
19. A finite angular-operation controller comprising a triplet-pair predicate, a scalar phase register, controlled positive angular contacts resolving a finite angular-contact spectrum, an addressed scalar phase operation and an inverse spectrum-resolution sequence, configured to reproduce a specified quadratic angular operation while leaving the non-triplet complement unchanged and retaining the accounted scratch, control, switching and inverse-operation resources.
20. The system of claim 1, further comprising a bounded-progress completion network, an independent final clock seal and a physical atomic counter-capture operation, with malformed, missing, duplicate and late acquired records retained as failed attempts rather than omitted from the operating denominator.

Claims 21–34. The following claims describe separately scoped representation, diagnostic method and apparatus embodiments.

21. Exchange-record representation method. A method of representing an optical-information exchange, comprising: specifying a joint preparation including a source, receiver, guide, clock records, references and retained apparatus recipients; specifying an ordered permitted update word and requested receiver records; retaining relative phase and record-overlap information affecting

those records; and merging descriptions only upon equality of the requested-record probabilities under every admitted future operation, or summing out a retained record after its last relevant contact, without treating the computational contraction as physical erasure.

22. Persistent-clock representation. The method of claim 21, wherein clock-conditioned interactions diagonal in three labels and cyclic clock advancement give $U^t=\Sigma_q|q+t\rangle\langle q|\otimes V_q(t)$, each site having a rank-at-most-three Gram kernel and its initial label retained through revisits, without replacing the collection of site records by three global features.

23. Separate coherence and population reduction. The method of claim 21, wherein a coherence query contracts relative amplitudes with retained reference records and a population query contracts full positive populations of the actual mixed preparation, without inferring the latter by squaring the former or discarding longitudinal or old-reference reactions.

24. Same-bank repeated-contact discriminator. A diagnostic method comprising: providing the full preparation of E18.1–E18.2; applying the existing clock-conditioned contact product F twice while retaining the same clock between contacts and making unselected operations identity in a separate no-transport word; addressing a distinct receiving instrument to the same source bank and matched mode; retaining joint phase-sensitive receiver records; and comparing their unconditioned mean with the persistent and independently renewed-clock predictions, wherein selective contacts and same-bank addressing satisfy AP-RCM.

25. Joint receiver access. The method of claim 24, wherein the receiving instrument uses the joint carrier/flag lowering transition and reacting phase stock of E18.3, applies relative phase and Uj, then records actual D_dark followed by D into distinct full C9 phase pointers on each rail, while retaining old references, flags, inactive stock and physical complements.

26. Full-domain frozen score. The method of claim 25, wherein labels 0…8 have score $[0,1,0,0,0,0,0,0,-1]$, the central pointer order is TOTAL_C,DARK_C,TOTAL_D,DARK_D, and $r=(\Sigma_i[s(T_C)-s(D_C)+s(T_D)-s(D_D)])/(n\sqrt{2})$, preserving every label and actual DARK with no clipping, selected-event normalization or rail averaging, the score having full-domain norm $2\sqrt{2}$.

27. Independent certificate. The method of claim 26, wherein a half-trace joint-preparation bound D_prep, unhalved-diamond complete-word bound δ_{word}, decoder-bias bound e_pointer and fixed-sample term e_stat satisfy $4\sqrt{2}\cdot D_{prep}+2\sqrt{2}\cdot\delta_{word}+e_{pointer}+e_{stat}\le 1/960$ through disjoint qualification evidence covering the full supplied domains and retained references.

28. Frozen bins and conservative failure. The method of claim 27, wherein the predictions are 1/32 and $\beta=(15-2\sqrt{3}-\sqrt{2}-\sqrt{6})/288$, separated by more than 1/240; a certified outward record interval wholly in [29/960,31/960] is persistent-compatible, wholly in [24/960,27/960] is renewal-not-excluded, disjoint from both rejects both frozen envelopes or posing/error conditions, and all other intervals are inconclusive; missing records, invalid certificates or failed required controls prevent passage.

29. Complete control schedule. The method of claim 28, further comprising the N,Q,V,I,S,O,R,M,H,Z rows of E18.6 with their specified targets, n fixed blocks in cyclically rotated table order, common controls evaluated against shared bands and model-specific Q,R,H controls against their respective hypothesis bands, with no control-derived rescaling of the primary result.

30. Accounted mimic and sham. The method of claim 29, wherein the mimic and sham satisfy the additional AP-M configuration, and the mimic exports the first clock and its correlations into independent stock and supplies fresh χ before the second contact without renewing order; the sham uses the same added stocks and two inverse-equivalent full swaps before the second contact; and every extra factor, recipient, control letter and protection obligation remains charged and retained.

31. Fixed statistical procedure. The method of claim 29, wherein e_stat=1/3840, n=3,774,873,600 per row, independently qualified bounded trials supply the stated Hoeffding family certificate, and any incomplete schedule or unqualified cross-trial dependence makes the comparison inconclusive without replacement trials or post-outcome changes to the budget.

32. Protected one-shot archive. The method of claim 24, wherein a matched independent archive receives the complete data and history by full SWAP, the executing program and active/off history remain separate, an externally commanded OFF-token SWAP occurs outside the A-controlled unitary program, and all admitted later contacts exclude the archived result before pointer extraction, without claiming reset, stock regeneration or autonomous repetition.

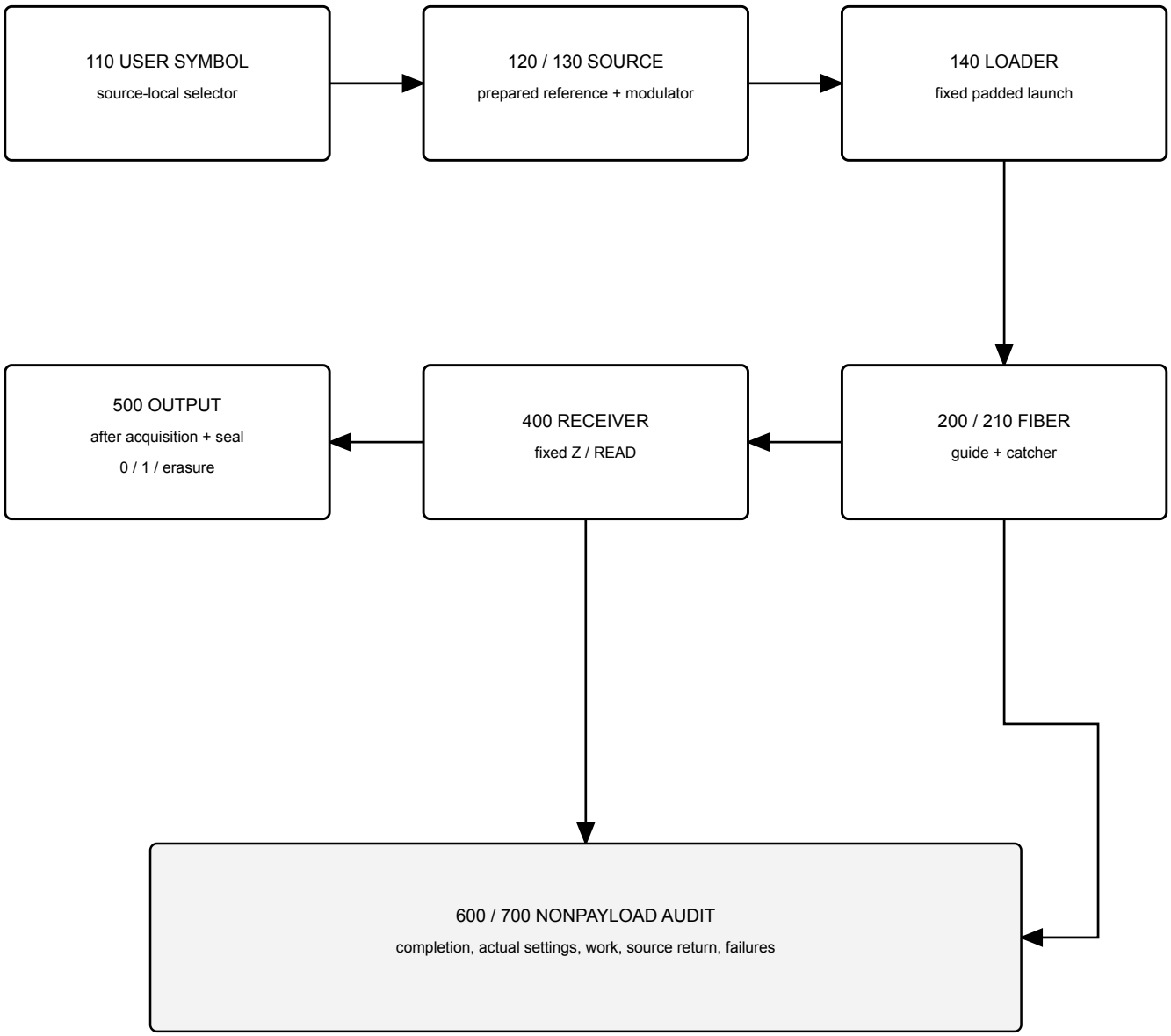
33. Replayable record reduction. A record-processing method for the protocol of claims 24–32, comprising linking a sealed schedule and qualification manifests to raw pointer observations and retained-unread custody records; preserving all scheduled slots including tombstones; accumulating exact integer scores and complete four-pointer histograms; checking marginal, count, identity and score consistency; determining exact or outward-interval bin and control dispositions; and emitting a reduction receipt linked by byte sizes and hashes to immutable originals and an append-only custody chain, wherein documentary validity is not substituted for physical certification.

34. A same-bank contact-memory diagnostic apparatus comprising: the two-rail source bank and full inventory of E18.1; persistent three-label scalar clocks controlling the full carrier/order contacts of E18.2; a finite controller specifying two same-clock contact products with identity spectators; a same-bank matched-mode receiver having distinct source/receiver stock and joint phase-reference access; four central C9 TOTAL/DARK phase pointers and the other retained receiving output pointers; a full data/history archive with separate program and OFF-token control; and a record processor implementing the frozen full-domain score, certificate, schedule and classification of E18.5–E18.9, wherein the contact itinerary and same-bank addressing satisfy AP-RCM and the required mimic/sham controls satisfy the additional AP-M configuration.

ABSTRACT

A fiber communications system encodes user information by a source-local angular operation and conveys the same signal organization through an actively controlled tree guide to a fixed receiver. Local scalar work pairs compensate site-tariff differences and retain a path-consistent work configuration. A receiver-occurrence decoder maps completed pointer states to symbols or erasures without consulting source settings or prepared-case identifiers. A separate acquisition subsystem retains completion, return, work and failed-attempt records. Source return does not erase measurement or thermal history; a finite fresh-unit embodiment supplies subsequent attempts explicitly. Common-coordinate matching, full-state positive contacts, retained references and distinct error metrics define the operating domain. Proposed guarded-contact and spectral-compilation embodiments further specify essential operations. Material implementation, physical input/output interfaces, calibration and reuse arrangements are selected for the operating domain.

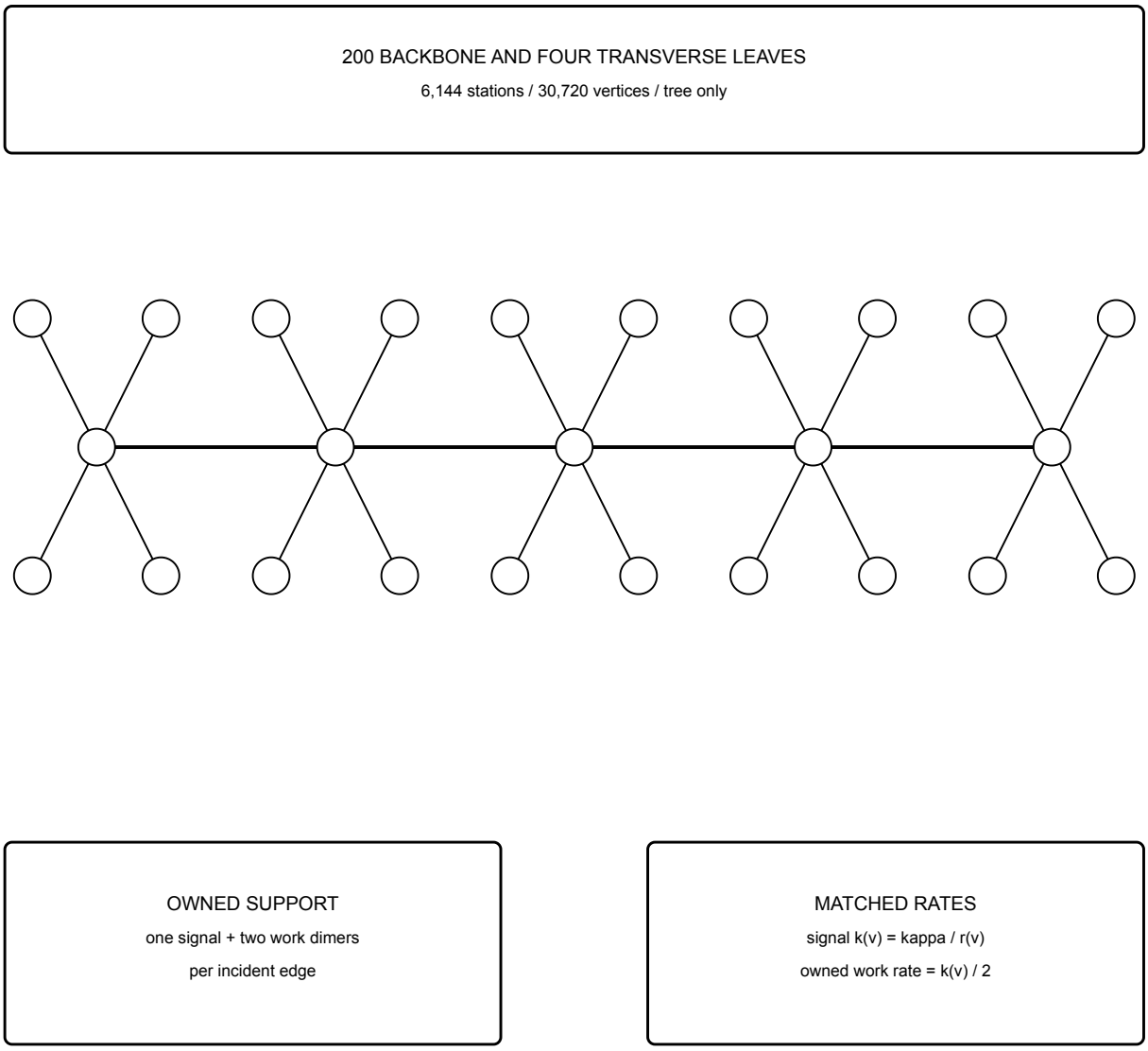
Complete source-to-output communications system and separate audit channel



The audit may inspect source settings; the payload decoder may not.

FIG. 1
Complete source-to-output communications system and separate audit channel.

Tree-shaped fiber, backbone, transverse leaves and owned work supports



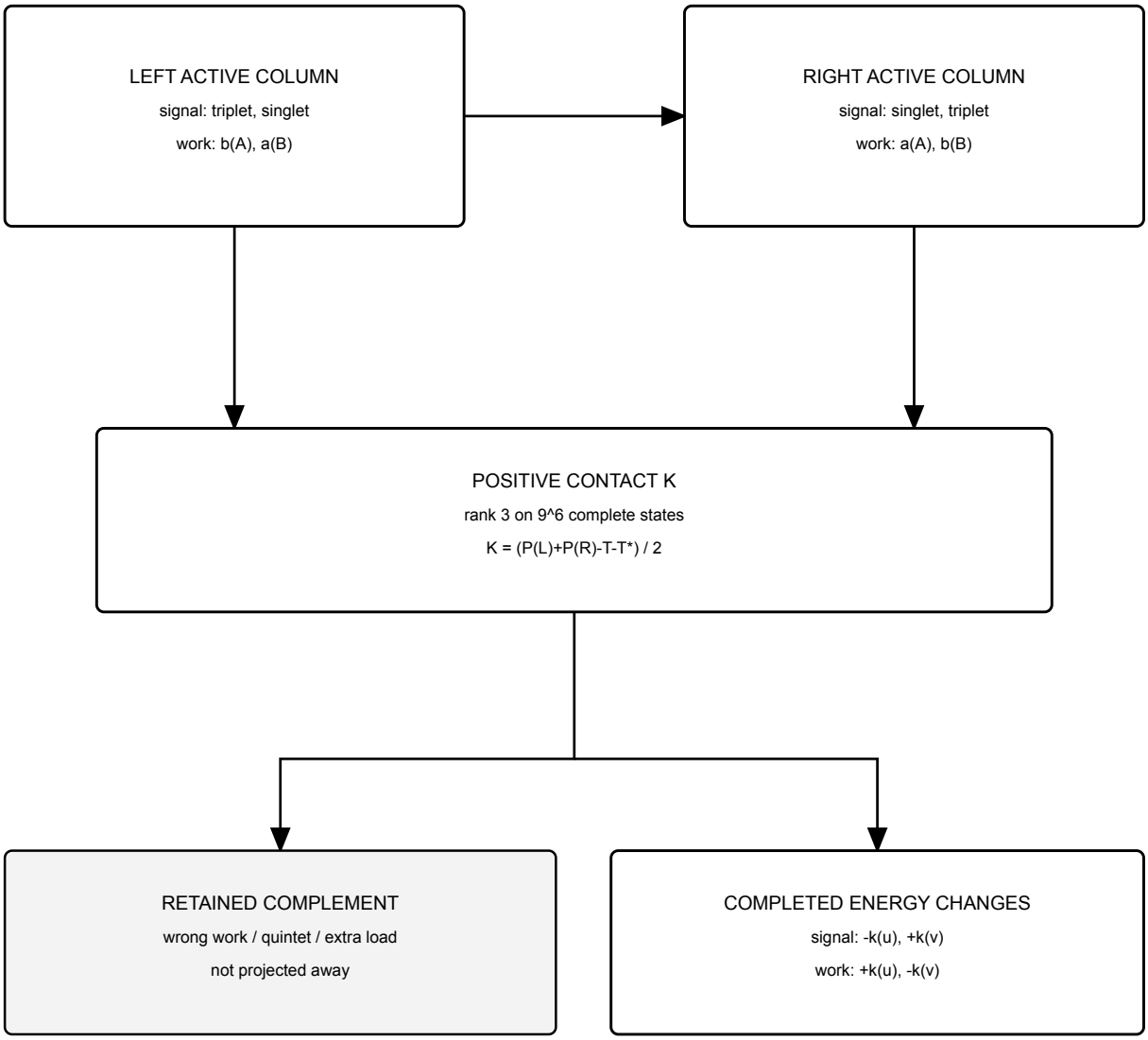
Source station 2,560 -> catcher station 3,584: 1,024 backbone edges.

No leaf-to-leaf axial edges. Diagram is schematic, not a calibrated fiber diameter.

FIG. 2

Tree-shaped fiber, backbone, transverse leaves and owned work supports.

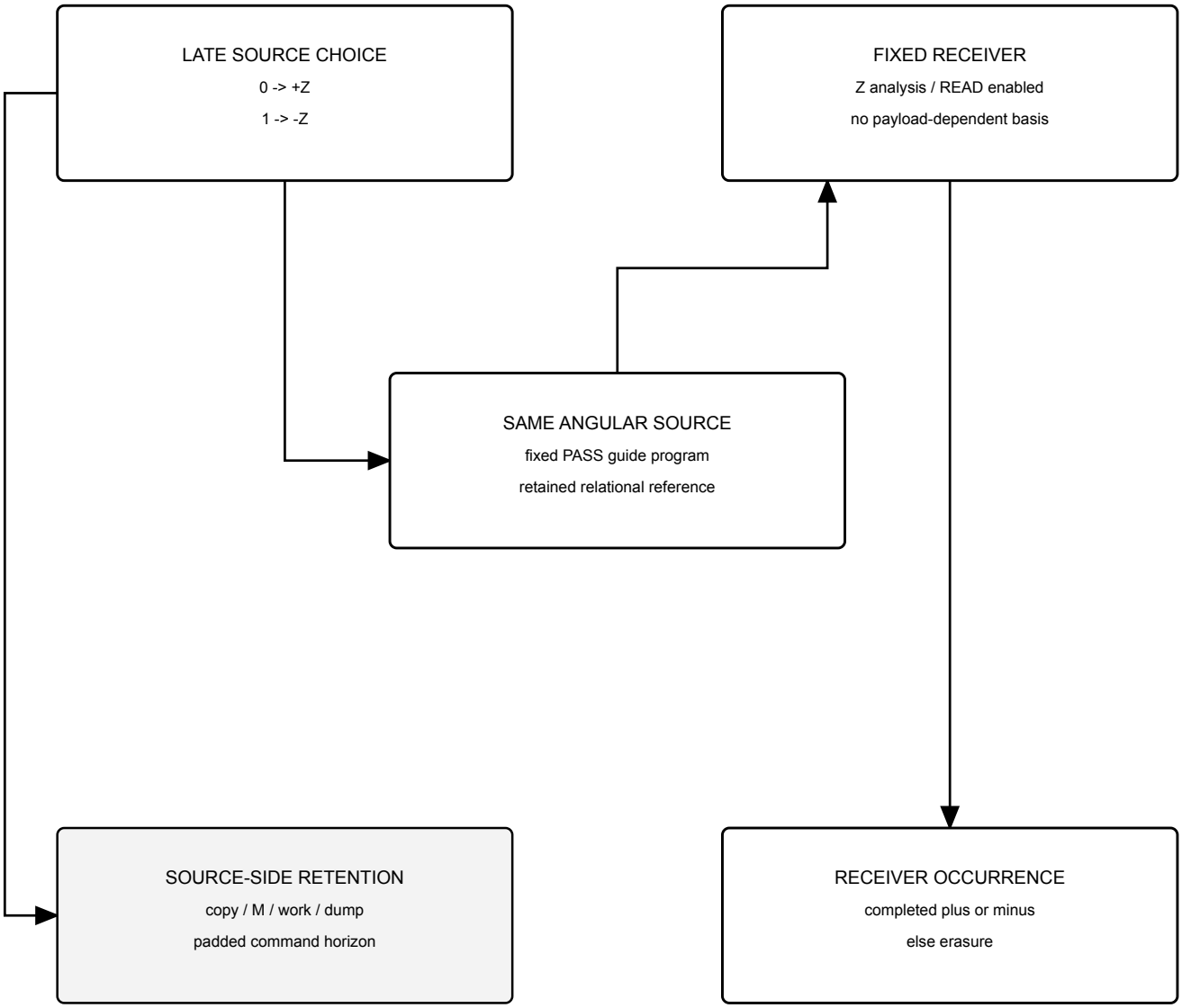
Work-compensated edge exchange and full-state complement



Both full work pairs are mounted on the arrival side; ownership determines their tariffs.

FIG. 3
Work-compensated edge exchange and full-state complement.

Source-local binary modulation with fixed receiver settings and retained reference



Endpoint n, q, reverse and blank controls remain fixed; their local phase is not remote payload.

FIG. 4
Source-local binary modulation with fixed receiver settings and retained reference.

Receiver-only decoding boundary and excluded source-setting metadata

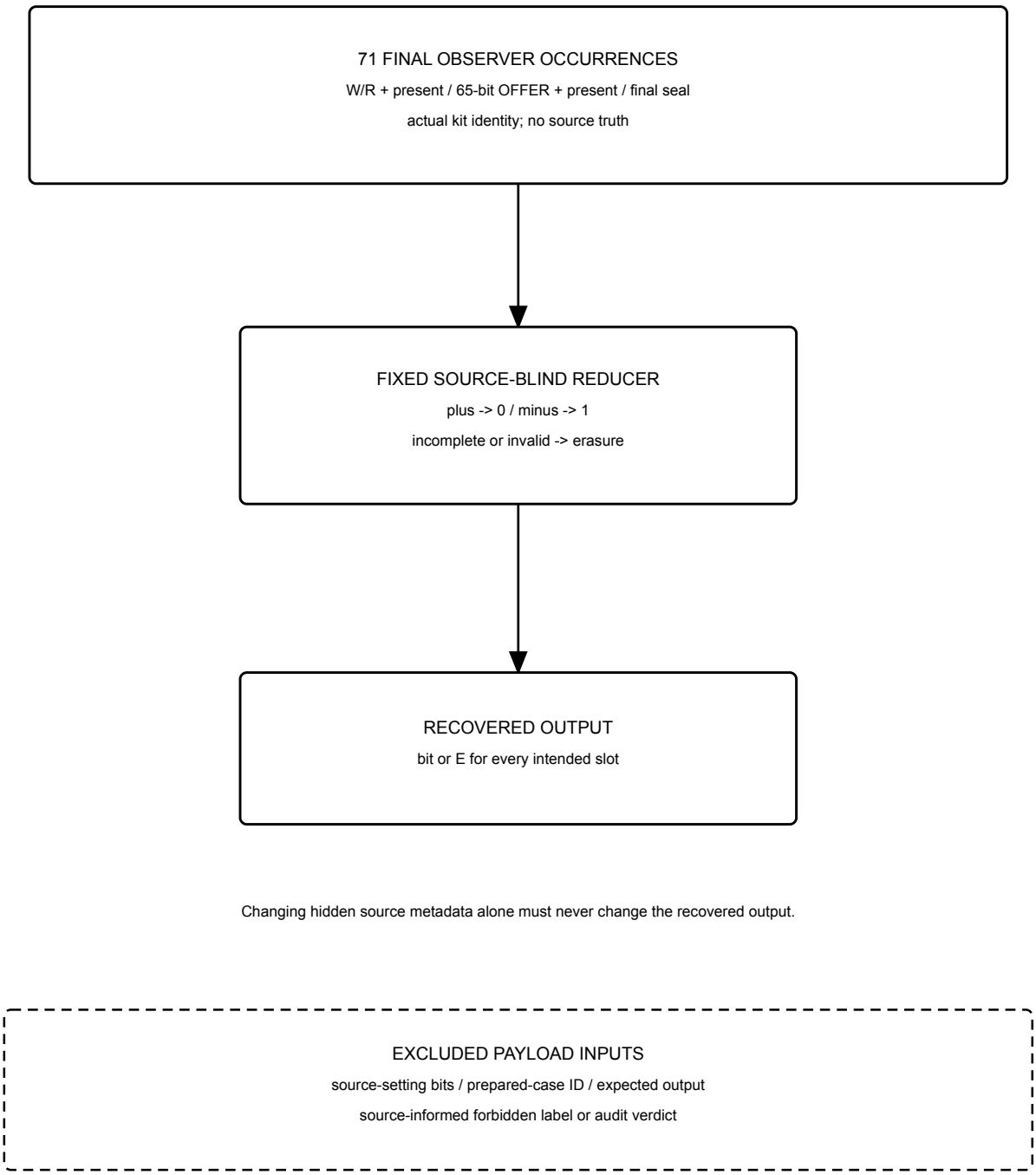
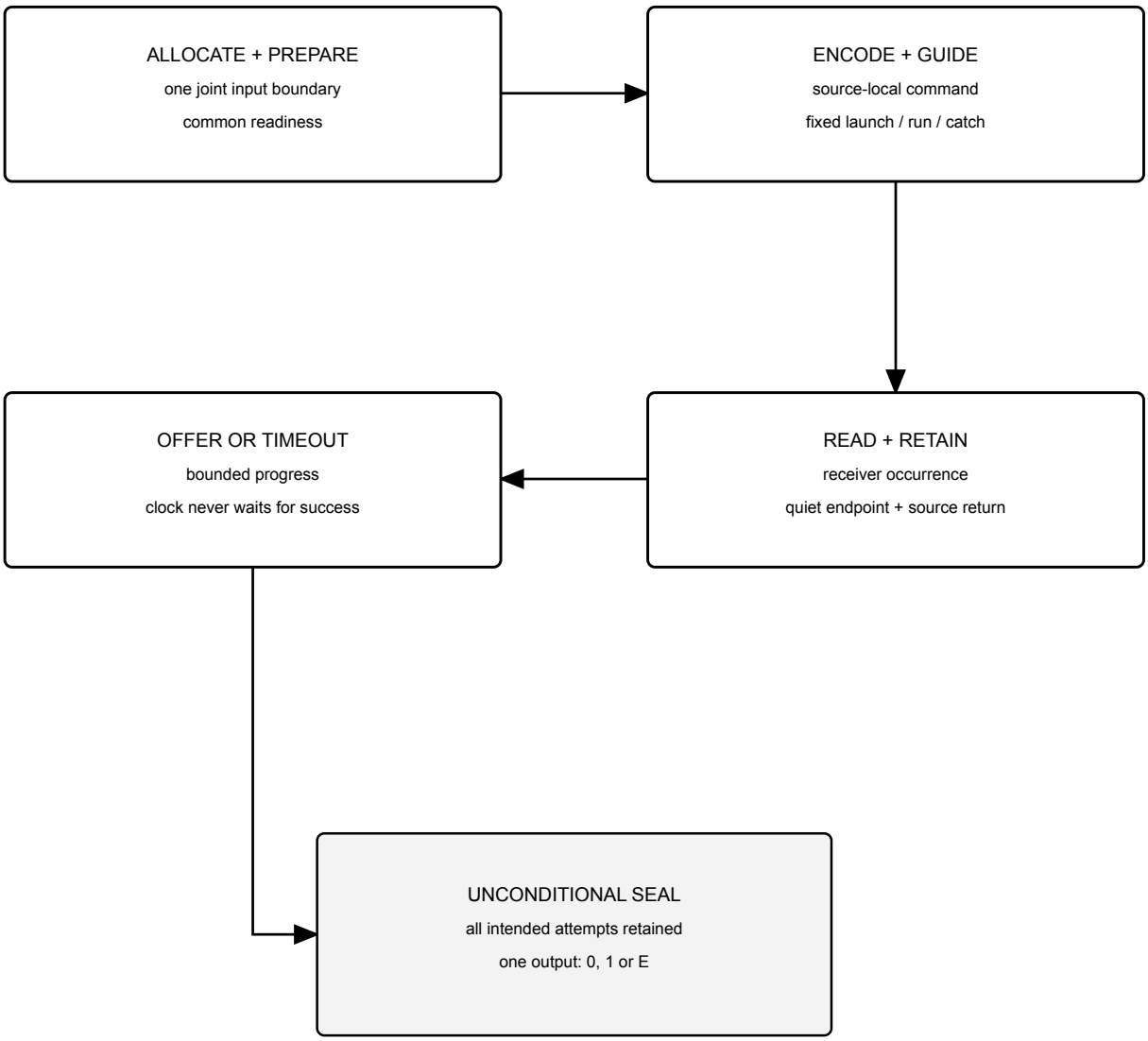


FIG. 5
Receiver-only decoding boundary and excluded source-setting metadata.

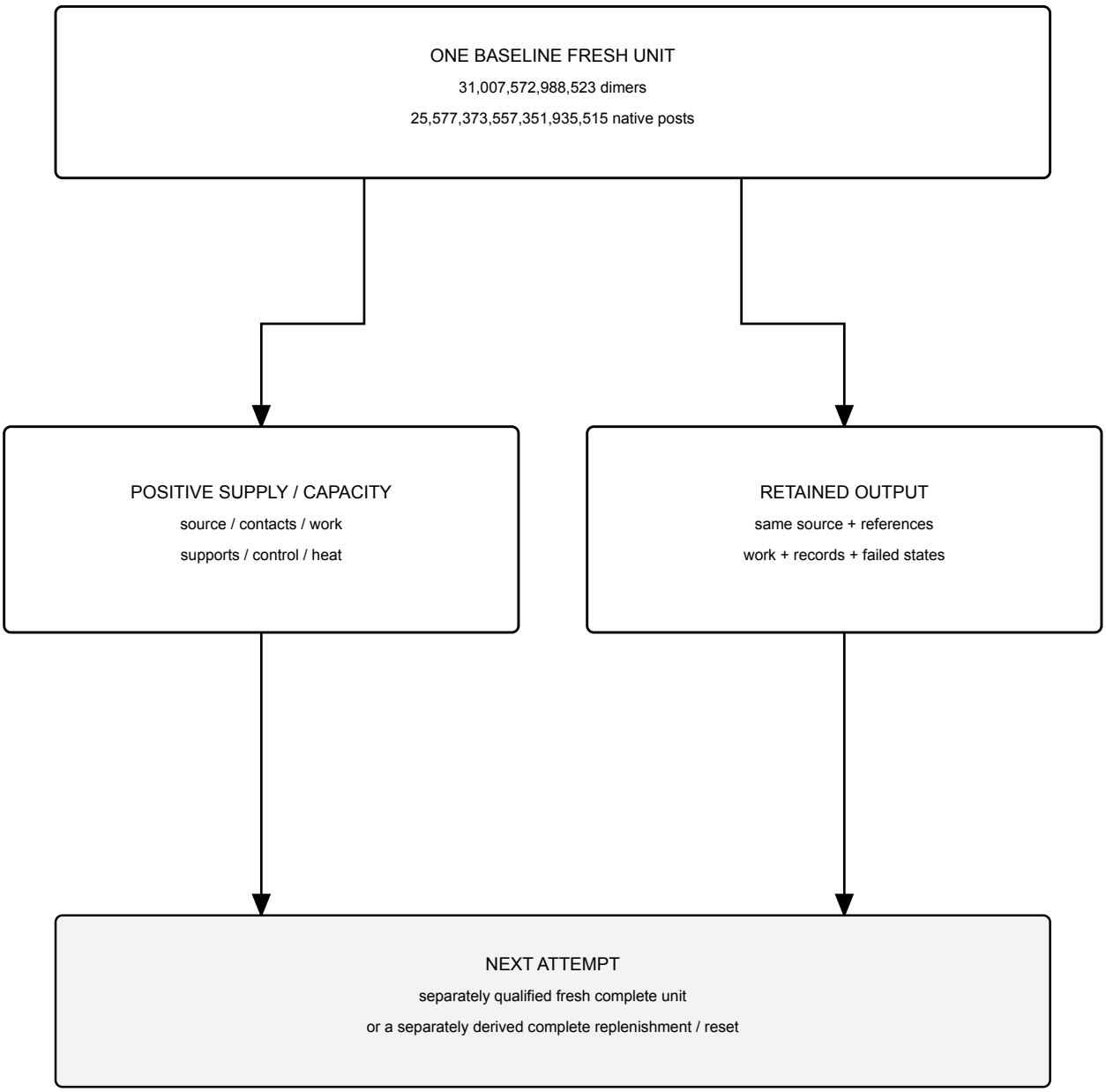
Finite attempted-symbol operation, timeout and unconditional retained record



Receiver-pointer time, observed-output time and complete-audit time are distinct.

FIG. 6
Finite attempted-symbol operation, timeout and unconditional retained record.

Complete positive resources, retained outputs and fresh-unit repetition



These are conditional sufficient upper accounts, not measured power, latency or fabrication cost.

FIG. 7
Complete positive resources, retained outputs and fresh-unit repetition.

Exact rate matching, guarded contact construction and material realization boundary



FIG. 8
Exact rate matching, guarded contact construction and material realization boundary.

Layered paper support and prospective qualification gates

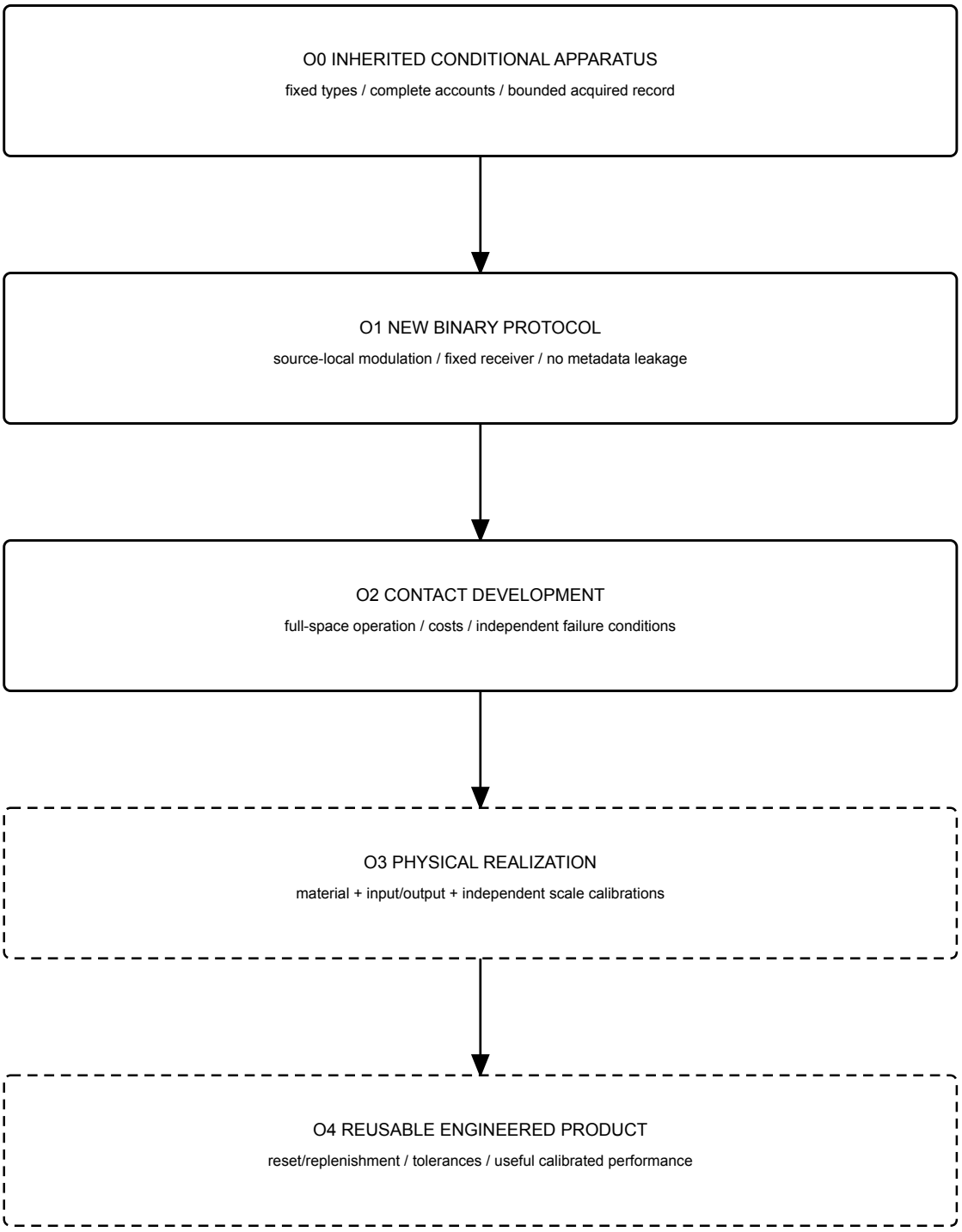
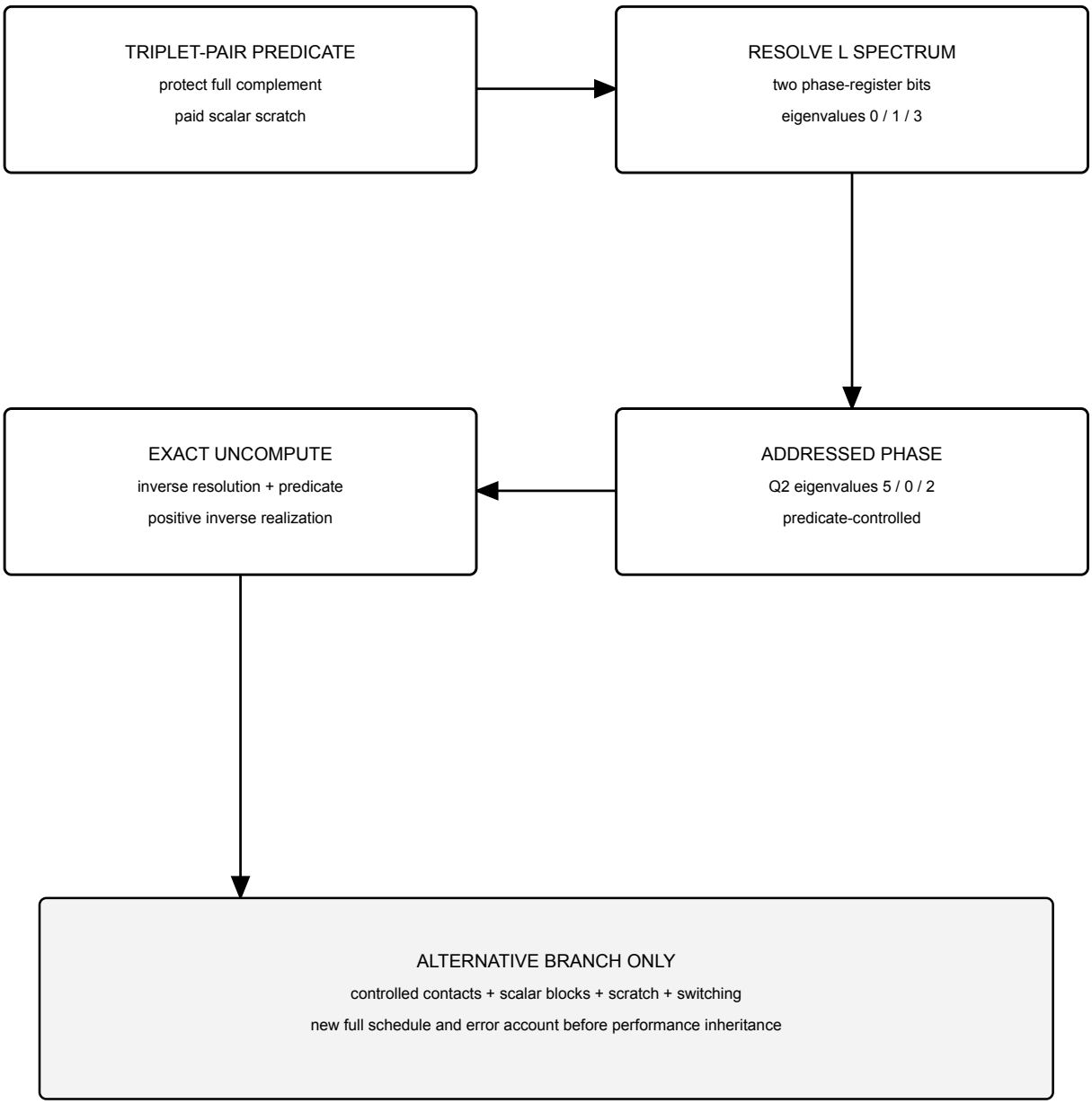


FIG. 9
Layered paper support and prospective qualification gates.

Alternative spectral compilation of the quadratic angular operation



This compiler reduces a prescribed primitive; it does not establish a physical material.

FIG. 10
Alternative spectral compilation of the quadratic angular operation.